

Electrostatics: Electrostatic potential and field due to discrete and continuous charge distributions, boundary condition, Energy for a charge distribution, Conductors and capacitors, Laplace's equation Image problem, Dielectric polarization, electric displacement vector, dielectric susceptibility, energy in dielectric systems

Textbooks:

1. **Introduction to Electrodynamics**, David J. Griffiths, Prentice Hall, 2007.
2. **Electromagnetic Field Theory Fundamentals**, Bhag Singh Guru, Huseyin R. Hiziroglu, Cambridge University Press; 2nd edition, 2009.

Electrostatics

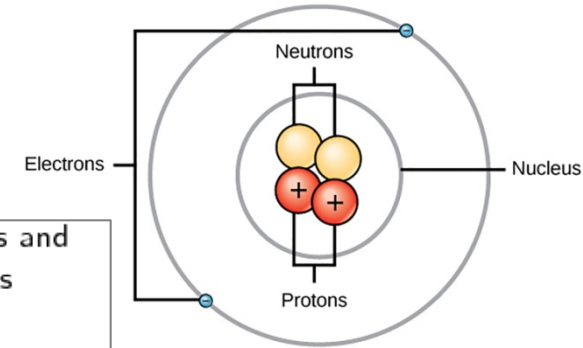
Introduction

- ❖ Electrostatics means *electric charges at rest* (*no moving charges*)
 - Objects become *charged by gaining or loosing electrons* by **two kind of charges**:

Positive: Shortage of Electrons

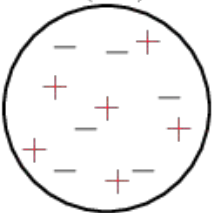
Negative: Excess of Electrons

Neutrons can't be transferred !



6 positive charges and
6 negative charges

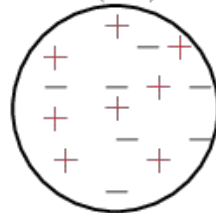
$$6 + (-6) = 0$$



There is zero net charge:
The object is neutral

8 positive charges and
6 negative charges

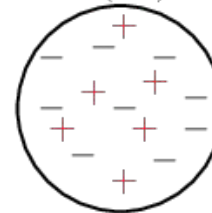
$$8 + (-6) = 2$$



The net charge is +2
The object is positively charged

6 positive charges and
9 negative charges

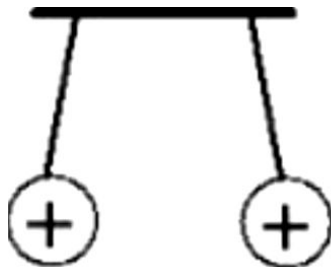
$$6 + (-9) = -3$$



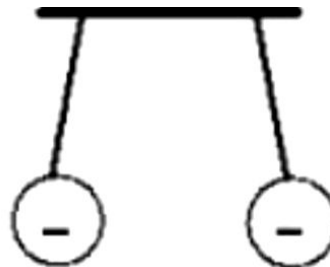
The net charge is -3
The object is negatively charged

- **Forces between charges:** *Same charges repel* each other

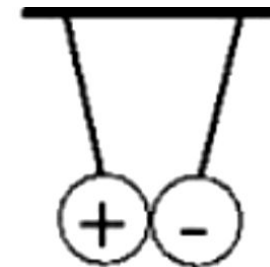
Opposite charges attract each other



Repel



Repel



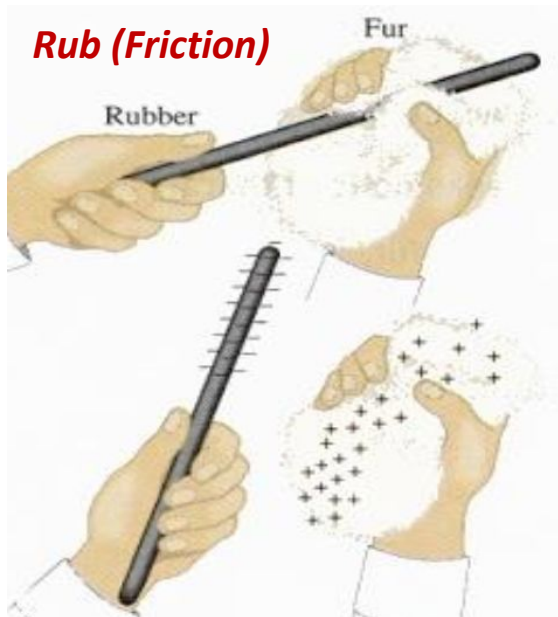
Attract

Electrostatics

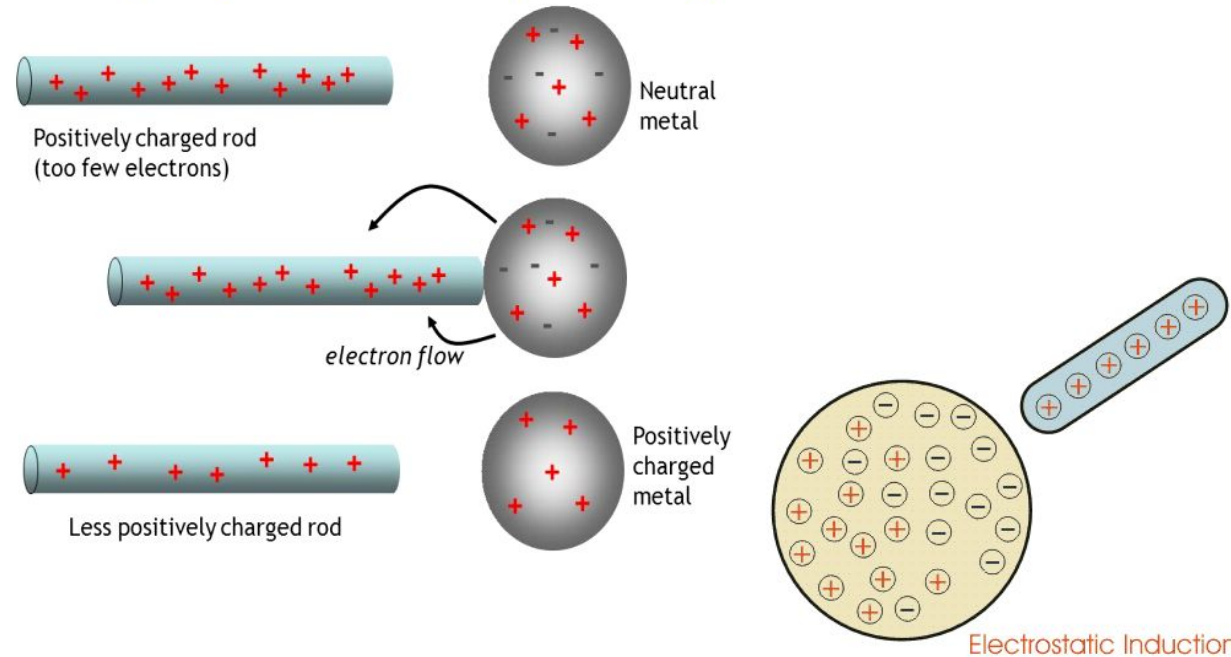
Introduction

Charging objects:

- Electrons are transferred, if **two objects are rubbed against each other** (*Friction*)
- Electrons are transferred, if **a charged object touches a neutral object** (*Touch*)



Charge by conduction (touching)



Polarization (*Induction*):

- Charged particles in a neutral object **rearrange**, if a charged object is **brought close to it** (Causes *temporary charge distribution*)

Conservation of Charge:

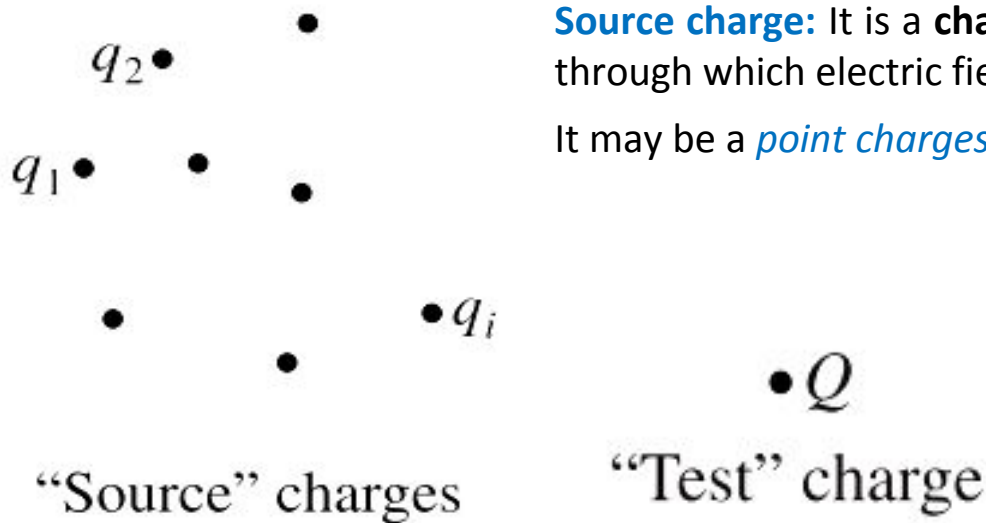
- Charge cannot be created or destroyed, *but only transferred from one object to another.*

Conductors, capacitors, dielectrics and their equation problems

Electrostatics

Electric field:

- The fundamental **problem in electrodynamics** expects to solve in this study.
- We have some electric charges, $q_1, q_2, q_3, \dots$ (call them **source charges**).
- Then, what force do they use on another charge, Q (call it as **test charge**)?



Source charge: It is a **charge to produce electric field** and is a **large charge** through which electric fields flow.

It may be a *point charges*, or a *continuous distribution of charges*.

Test charge: It is a **point charge that does not disturb any field**.

It is a *small electro positive charge* used for experiments.

- The **positions of the source charges** are given as *functions of time*, then the **path of the test particle** is to be calculated.
- In general, both the **source charges** and the **test charge** are **in motion**.

Electrostatics

Electric field

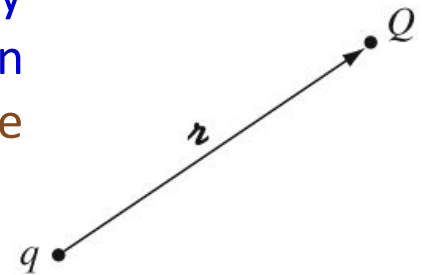
- The **solution to this problem** is facilitated by the principle of **superposition**, which states that the *interaction between any two charges is completely unaffected by the presence of others.*

Superposition: The ability of a quantum system to be in multiple states at the same time until it is measured.

- This means that to determine the force on 'Q', we can first compute the force $\mathbf{F}_1$, due to q_1 alone, then we compute the force $\mathbf{F}_2$, due to q_2 alone and so on.
- Finally, **we take the vector sum of all these individual forces**: $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$

□ Thus, if we can find the force on 'Q' due to a single source charge 'q' then **we are able to determine the force.**

□ Another important thing to notice here that the **force not only depends on charge 'Q' and the separation distance 'r' between the charges**, it also depends on both their **velocities** and on the **acceleration** of q.



- What is the force on Q due to q?**
- We shall consider the **special case of electrostatics** in which all the **source charges are stationary** (although the “test charge” may be moving).

Coulomb's law:

What is the force on a test charge “Q” due to a single point charge q, that is at rest a distance ‘r’ away?

The answer can be by Coulomb's law: $\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$

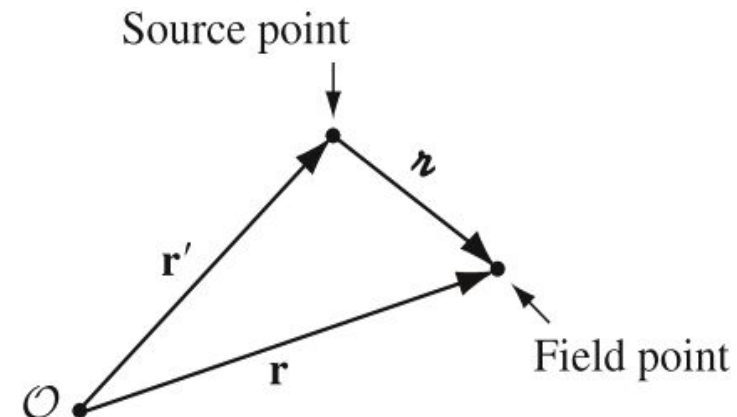
Here, the constant ϵ_0 is called the **permittivity of free space**. In SI units, where force is in newtons (N), distance in meters (m), and charge in coulombs (C)

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

The **force is proportional** to the **product of the charges** and **inversely proportional** to the **square of the separation distance**.

□ $\mathbf{r}$ is the separation vector from $\mathbf{r}'$ (the location of q) to $\mathbf{r}$ (the location of Q): $\mathbf{r} = \mathbf{r} - \mathbf{r}'$

r is its magnitude, and $\hat{\mathbf{r}}$ is its direction.

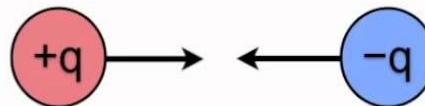


- The **force** points along the line **from q to Q**.
- It is **repulsive** if q and Q have the **same sign**, and **attractive** if their **signs are opposite**.
- **Coulomb's law and the principle of superposition** constitute the **physical input** for electrostatics.

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$$

Coulomb's law

The closer two charges are, the stronger the force between them



Electric field

Coulomb's Law-EF:

- If we have several point charges $q_1, q_2, \dots, q_n$, at distances $r_1, r_2, \dots, r_n$ from Q , the total force on Q is evidently.

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}}$$

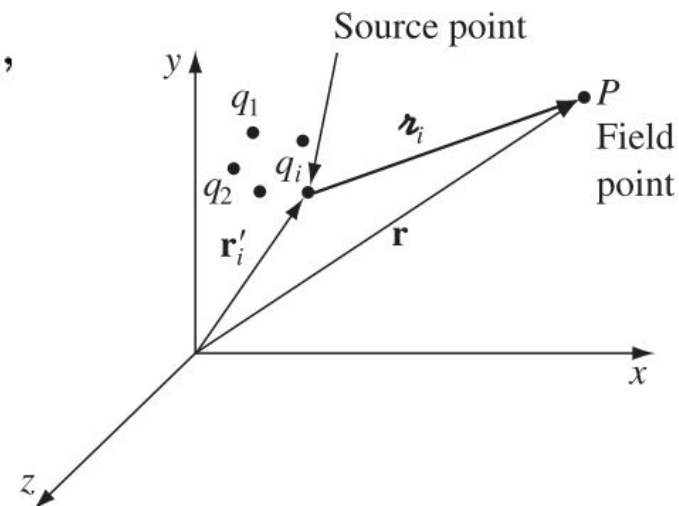
$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2 Q}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right)$$

$$= \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right),$$

or $\mathbf{F} = QE$

,

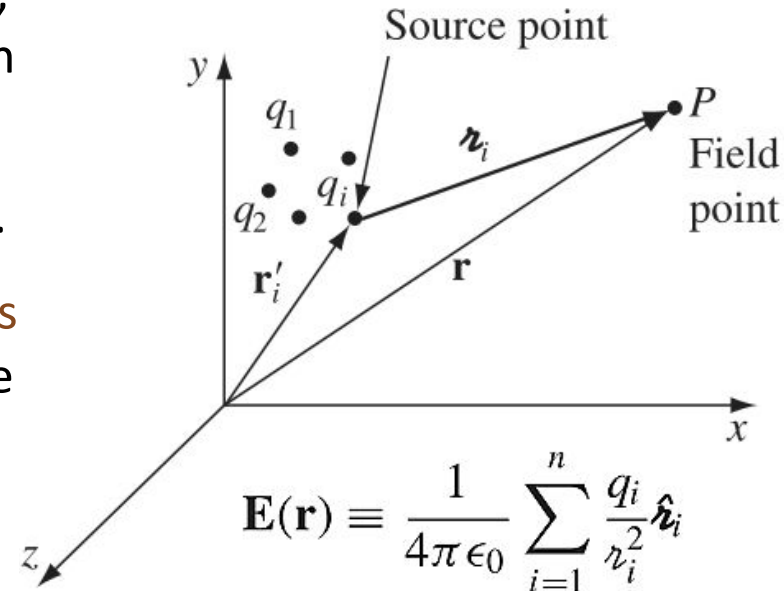
where, $\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$



Here, $\mathbf{E}$ is called the **electric field** of the source charges.

Electric field

- Note that “E” is a function of position ($\mathbf{r}$), because the **separation vectors** $\mathbf{r}_i$ depend on the location of the **field point** ‘P’.
- But, it makes **no reference** to the **test charge** Q.
- The **electric field** is a **vector quantity** that **varies from point to point** and is determined by the configuration of **source charges** ($q_1, q_2, \dots$).



- Physically, $\mathbf{E}(\mathbf{r})$ is the **force per unit charge** that would be exerted on a *test charge*, if you were to place one at P.

What exactly is an electric field?

Maxwell himself came to believe that **electric** and **magnetic fields** are **stresses** and **strains** in an **invisible primordial** jellylike “**ether**”. (*a theoretical universal substance*)

- **Special relativity** has forced us to abandon the idea of ether, and with it Maxwell’s mechanical interpretation of electromagnetic fields.

Electric field

Problem: Find the electric field a distance ' z ' above the midpoint between two equal charges (q), a distance ' d ' apart?

Solution: Let $\mathbf{E}_1$ be the field of the left charge alone, and $\mathbf{E}_2$ that of the right charge alone.

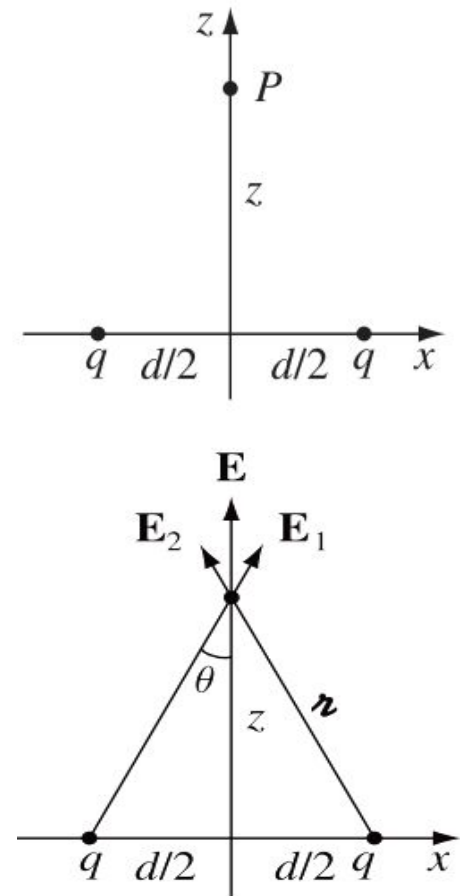
Adding them (vectorially), the horizontal components cancel and the vertical components plotted:

$$E_z = 2 \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cos \theta$$

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

Here $r = \sqrt{z^2 + (d/2)^2}$ and $\cos \theta = z/r$, so

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2qz}{[z^2 + (d/2)^2]^{3/2}} \hat{\mathbf{z}}.$$



Check: When $z \gg d$ you're so far away that it just looks like a single charge $2q$, so the field should reduce to $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2q}{z^2} \hat{\mathbf{z}}$. And it *does* (just set $d \rightarrow 0$ in the formula).

Electric field

Continuous charge distribution:

□ Our definition of the electric field assumes that the **source of the field is a collection of discrete point charges q_i** .

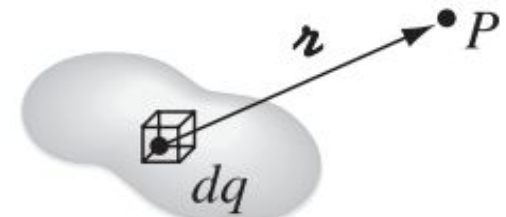
□ The **charge is distributed continuously over some region**, the **sum becomes an integral**.

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

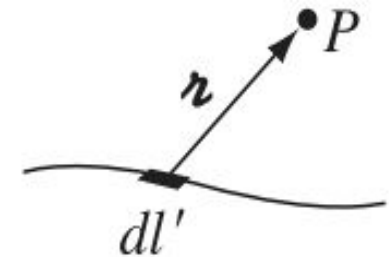
$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq$$

□ If the charge is **spread out along a line**, with charge-per-unit-length λ , then, **$dq = \lambda dl'$**

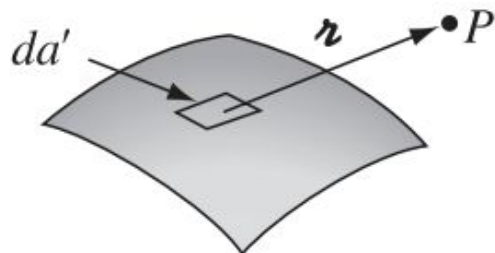
Where, dl' is an element of length along the line



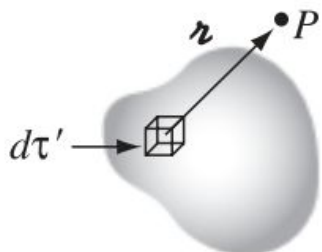
Continuous distribution



Line charge, λ



Surface charge, σ



Volume charge, ρ

□ If the charge is **smeared out over a surface**, with charge-per-unit-area σ , then **$dq = \sigma da'$**

where da' is an element of area on the surface

□ If the **charge fills a volume**, with charge-per-unit-volume ρ , then **$dq = \rho d\tau'$**

where, $d\tau'$ is an element of volume

Continuous charge distribution

□ Hence, in general

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} dq$$

$$dq \rightarrow \lambda dl' \sim \sigma da' \sim \rho d\tau'$$

□ Thus, the electric field of a **line charge**: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2} \hat{\mathbf{r}} dl'$

□ For a **surface charge**: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2} \hat{\mathbf{r}} da'$

□ For a **volume charge**: $\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'$

The last (volume) equation itself is often referred to as “**Coulomb’s law**”.

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r^2} \hat{\mathbf{r}} d\tau'$$

Continuous charge distribution

Problem: Find the electric field a distance z above the midpoint of a straight line segment of length $2L$ that carries a uniform line charge λ ?

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2} \hat{\mathbf{r}} dl'$$

Solution: $\mathbf{r} = z \hat{\mathbf{z}}, \quad \mathbf{r}' = x \hat{\mathbf{x}}, \quad dl' = dx;$

$$\mathbf{r} = \mathbf{r} - \mathbf{r}' = z \hat{\mathbf{z}} - x \hat{\mathbf{x}}, \quad r = \sqrt{z^2 + x^2}, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r} = \frac{z \hat{\mathbf{z}} - x \hat{\mathbf{x}}}{\sqrt{z^2 + x^2}}.$$

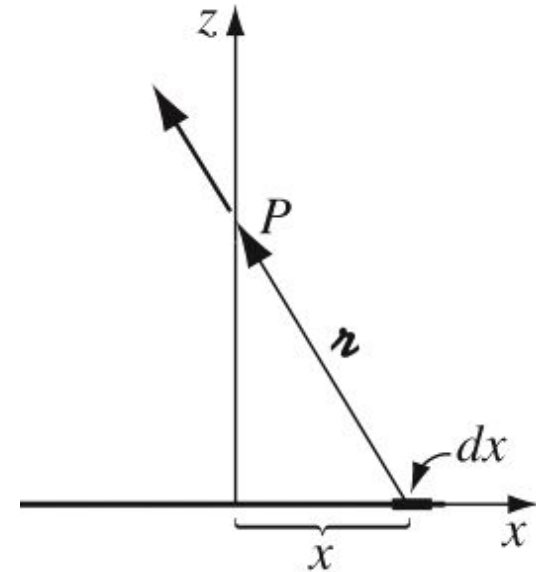
$$\begin{aligned} \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \int_{-L}^L \frac{\lambda}{z^2 + x^2} \frac{z \hat{\mathbf{z}} - x \hat{\mathbf{x}}}{\sqrt{z^2 + x^2}} dx \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z \hat{\mathbf{z}} \int_{-L}^L \frac{1}{(z^2 + x^2)^{3/2}} dx - \hat{\mathbf{x}} \int_{-L}^L \frac{x}{(z^2 + x^2)^{3/2}} dx \right] \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[z \hat{\mathbf{z}} \left(\frac{x}{z^2 \sqrt{z^2 + x^2}} \right) \Big|_{-L}^L - \hat{\mathbf{x}} \left(-\frac{1}{\sqrt{z^2 + x^2}} \right) \Big|_{-L}^L \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z \sqrt{z^2 + L^2}} \hat{\mathbf{z}}. \end{aligned}$$

For points far from the line ($z \gg L$),

$$E \cong \frac{1}{4\pi\epsilon_0} \frac{2\lambda L}{z^2}.$$

This makes sense: From far away the line looks like a point charge $q = 2\lambda L$. In the limit $L \rightarrow \infty$, on the other hand, we obtain the field of an infinite straight wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}$$



Field lines

- Let's begin with the simplest possible case: a **single point charge q** , situated at the origin.

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

- The **field falls off like $1/r^2$** , the **vectors get shorter** if it goes further **away from the origin**, then they **always point radially outward**.

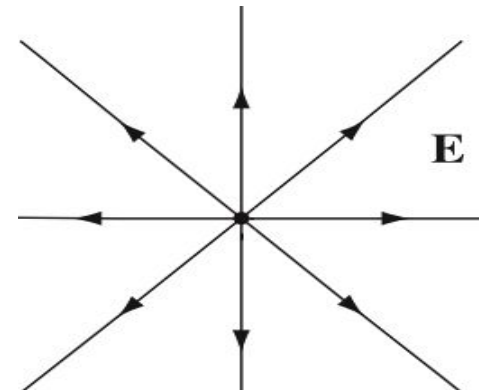
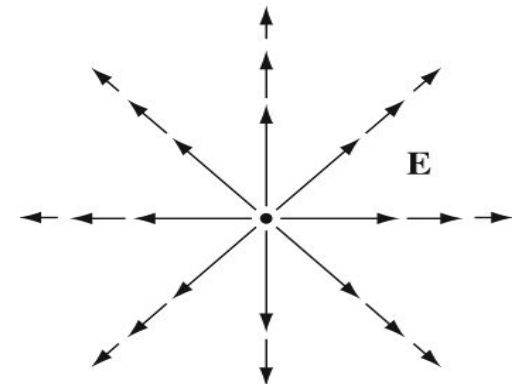
- The **magnitude of the field** is indicated by the **density of the field lines**:

- It is **strong near the center**, where the field lines are close together
- It is **weak**, where they are relatively far apart.

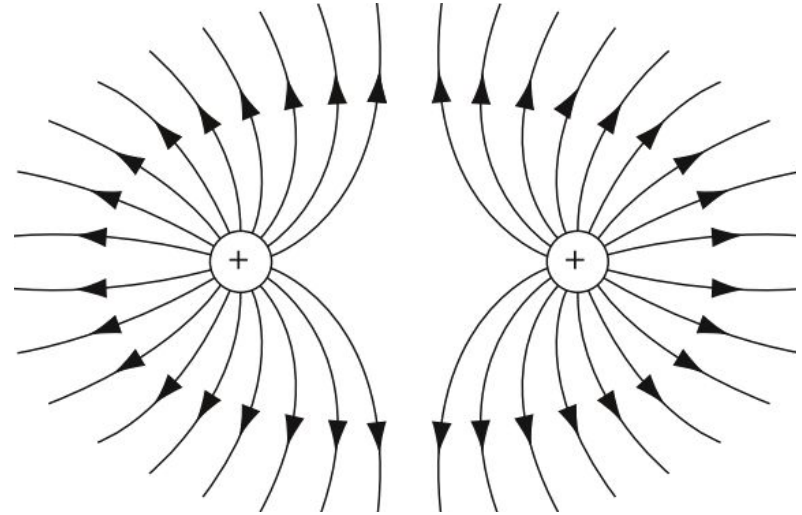
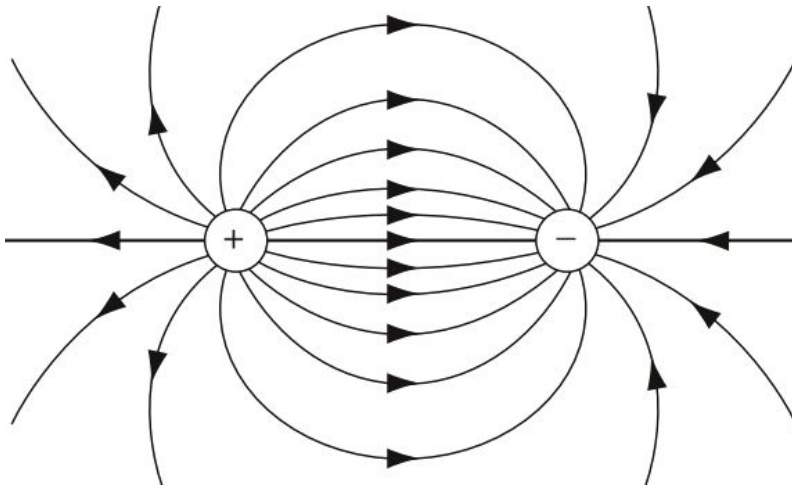
- By **drawing the lines** in the neighborhood of each charge, and then **connect them up or extend them to infinity**, then the **flux of $\mathbf{E}$ through a surface S** :

$$\Phi_E \equiv \int_S \mathbf{E} \cdot d\mathbf{a}$$

It is a measure of the “**number of field lines**” passing through S .



Field lines and flux



- For a **given sampling rate**, the **flux is proportional to the number of lines drawn**, because the *field strength is proportional to the density of field lines*.
- ✓ Hence, **$\mathbf{E} \cdot d\mathbf{a}$** is proportional to the number of lines passing through the infinitesimal area **$d\mathbf{a}$** .

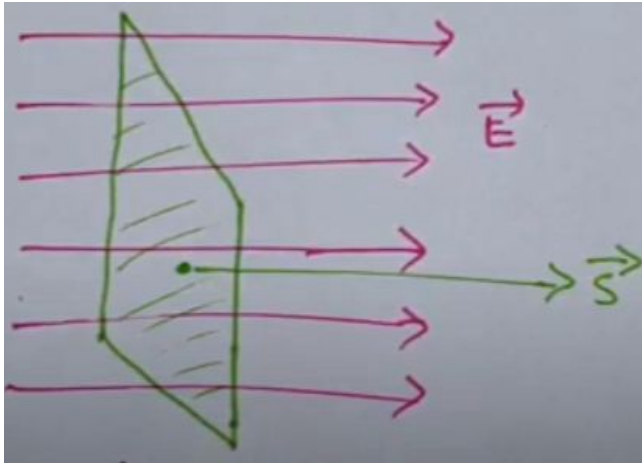
$$\Phi_E \equiv \int_S \mathbf{E} \cdot d\mathbf{a}$$

This suggests that the **flux through any closed surface** is a measure of the **total charge inside**.

Electric flux

□ Electric field lines passing through any surface is **Electric Flux**. $\Phi_E \equiv \int_S \mathbf{E} \cdot d\mathbf{a}$

If, electric field is uniform:

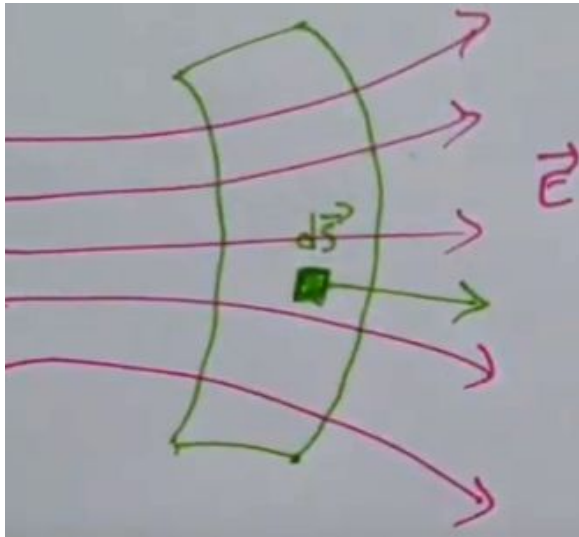


Electric Flux:

$$\Phi_E = \vec{E} \cdot \vec{S} = E \cdot S \cos \theta$$

if $\theta = 0$, $\Phi_E = E \cdot S$

If, electric field is non-uniform:



For small surface $d\vec{S}$, flux will be

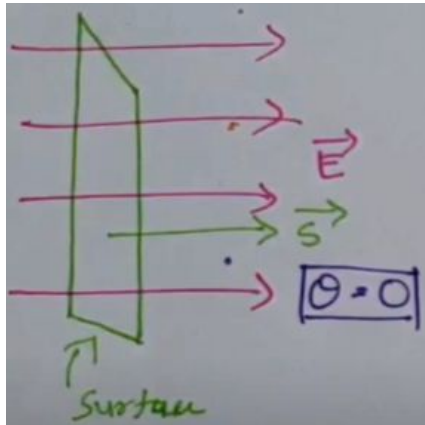
$$d\Phi_E = \vec{E} \cdot d\vec{S} = E \cdot S \cos \theta$$

For total electric flux,

$$\Phi_E = \int d\Phi_E = \int \vec{E} \cdot d\vec{S}$$

Electric flux

□ The electric flux at different cases:



Case-I: If, $\theta = 0^\circ$

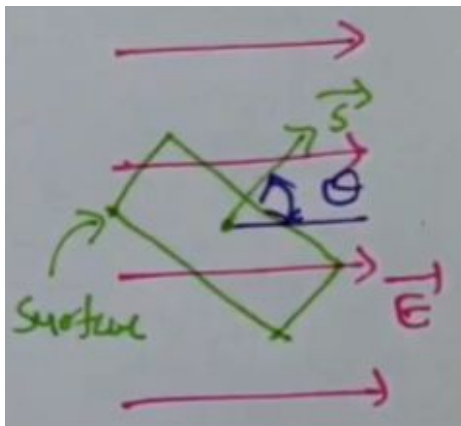
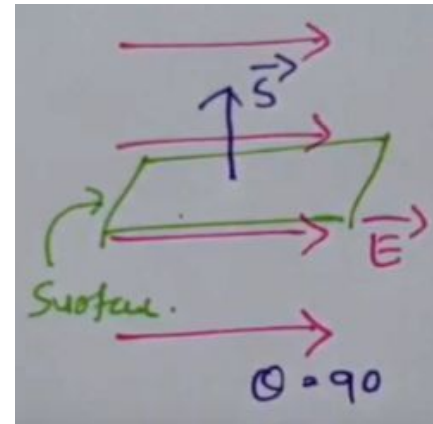
$$\Phi_E = \vec{E} \cdot \vec{S} = E.S \cos \theta$$

if $\theta = 0$, $\Phi_E = E.S$

Case-II: If, $\theta = 90^\circ$

$$\Phi_E = \vec{E} \cdot \vec{S} = E.S \cos \theta$$

if $\theta = 90^\circ$, $\Phi_E = 0$

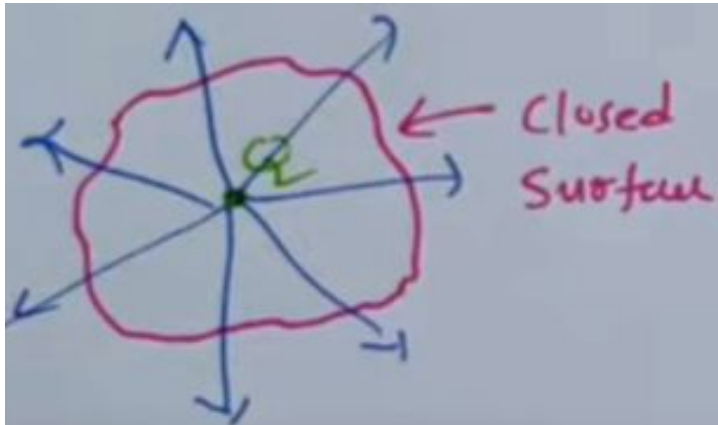


Case-III: If, $0^\circ < \theta < 90^\circ$

$$\begin{aligned} \Phi_E &= \vec{E} \cdot \vec{S} \\ &= E.S \cos \theta \end{aligned}$$

Gauss's law

- Gauss's law states that total **electric flux passing through any closed surface** is equal to **enclosed charge by the surface divided by ϵ_0** (permittivity of free space).



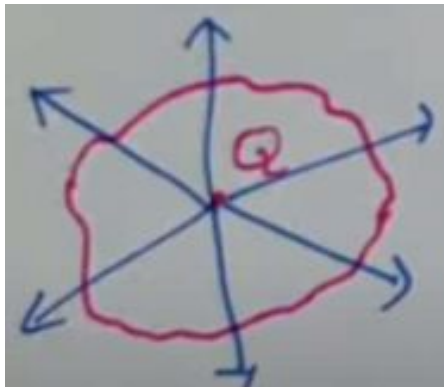
Permittivity of free space is a **physical constant** that reflects the ability of electrical fields to pass by a classical vacuum.

$$\epsilon_0 = 1/\mu_0 c^2 = 1/(4\pi \times 10^{-7} \text{ N/A}^2)(3 \times 10^8 \text{ m/s}^2)^2 = 8.854 \times 10^{-12} \text{ C/Nm}^2$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

Gauss's Law at different cases (Electric Flux):

Case-I



$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{s} \\ &= \frac{Q_{enc}}{\epsilon_0}\end{aligned}$$

Case-II

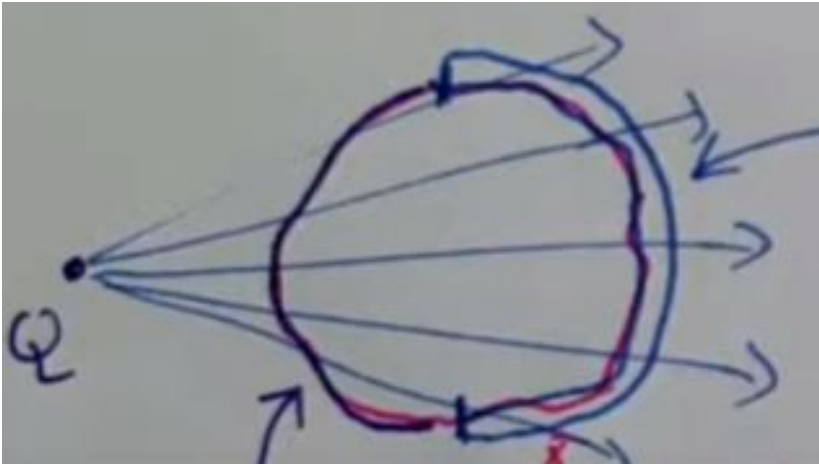


$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{s} \\ &= \frac{-Q_{enc}}{\epsilon_0}\end{aligned}$$

- ✓ If **field lines is leaving** the enclosed surface then electric **flux is positive**.
- ✓ If **field lines is entering** inside the enclosed surface then electric **flux is negative**.

Gauss's law

Case-III



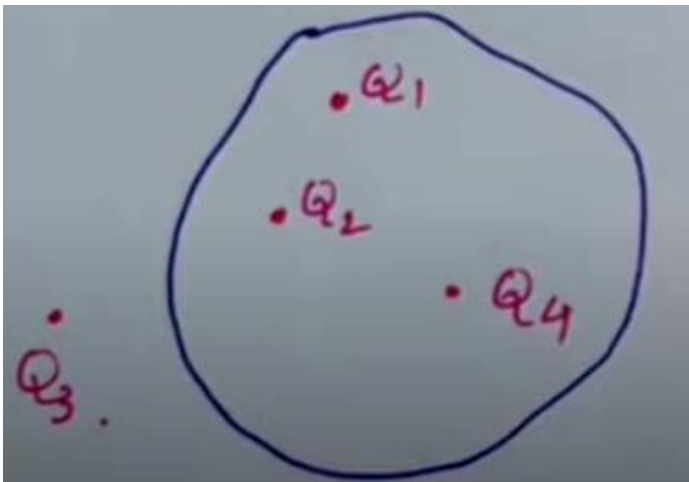
Flux is leaving outside of the surface

Hence, flux is **positive (+_{ve})**

Flux is entering inside of the surface
Hence, flux is **negative (-_{ve})**

Therefore, the **total flux** will be “zero” (0).

Example:



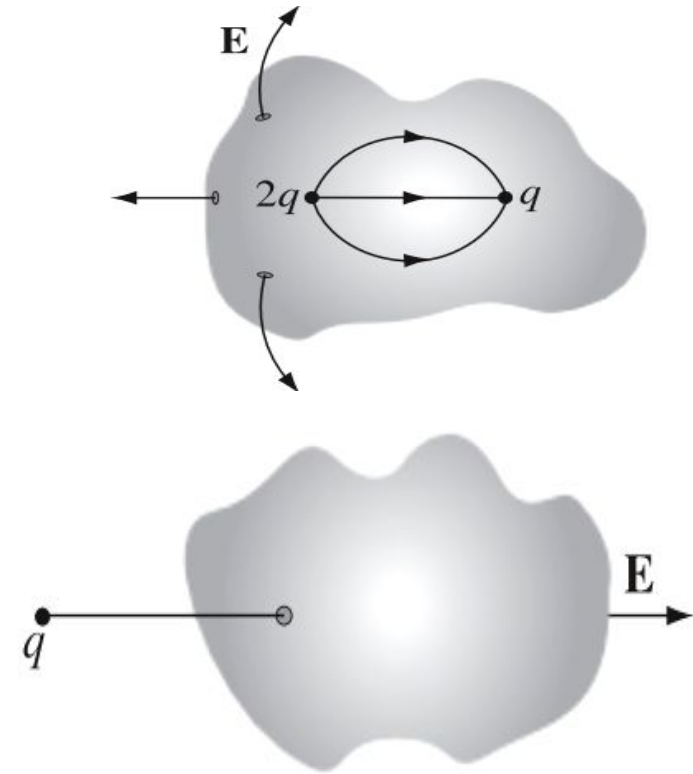
$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{s} \\ &= \frac{Q_{enc}}{\epsilon_0} = \left(\frac{Q_1 + Q_2 + Q_4}{\epsilon_0} \right)\end{aligned}$$

$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{s} \\ &= Q_{enc} = Q_1 + Q_2 + Q_4\end{aligned}$$

Gauss's law

- For the field lines that **originate on a positive charge** must **either pass out through the surface** or else **terminate on a negative charge inside**.
- On the other hand, **a charge outside the surface** **can't contribute anything to the total flux**, since *its field lines pass in out side*.

This is the essence of **Gauss's law**



- ✓ In the case of a **point charge 'q'** at the origin, the **flux of E** through a **spherical surface** of radius **r** is:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \int \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \hat{\mathbf{r}} \right) \cdot (r^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) = \frac{1}{\epsilon_0} q$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

Gauss's law

□ The flux through any surface enclosing the charge is q/ϵ_0 .

$$\Phi_E = \frac{Q_{enc}}{\epsilon_0}$$

□ Now suppose that instead of a single charge at the origin, we have a bunch of charges scattered about.

□ According to the **principle of superposition**, the **total field** is the **(vector) sum of all the individual fields**:

$$\mathbf{E} = \sum_{i=1}^n \mathbf{E}_i$$

□ The **flux through a surface that encloses them all is**

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} q$$

$$\oint \mathbf{E} \cdot d\mathbf{a} = \sum_{i=1}^n \left(\oint \mathbf{E}_i \cdot d\mathbf{a} \right) = \sum_{i=1}^n \left(\frac{1}{\epsilon_0} q_i \right)$$

□ For **any closed surface**:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{enc}$$

where, Q_{enc} is the **total charge enclosed** within the surface.

This is the **quantitative statement of Gauss's law**.

Gauss's law

- Gauss's law is an integral equation, but we can easily turn it into a differential one, by applying the **divergence theorem**:

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{E}) d\tau$$

- Rewriting Q_{enc} in terms of the charge density ρ , we have $Q_{\text{enc}} = \int_V \rho d\tau$

- Hence, Gauss's law becomes: $\int_V (\nabla \cdot \mathbf{E}) d\tau = \int_V \left(\frac{\rho}{\epsilon_0} \right) d\tau$

$$\Phi_E = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

- Furthermore, this holds for any volume, the integrands must be equal.

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$$

This is **Gauss's law in differential form**.

Gauss's law is always *true*, **but it is not always useful**.

→ **Symmetry is crucial** to this application of Gauss's law.

1. **Spherical symmetry**. Make your Gaussian surface a concentric sphere.
2. **Cylindrical symmetry**. Make your Gaussian surface a coaxial cylinder.
3. **Plane symmetry**. Use a Gaussian "pillbox," which straddles the surface.

Gauss's law

Applications of Gauss's Law

Problem: Find the field outside a uniformly charged solid sphere of radius R and total charge q ?

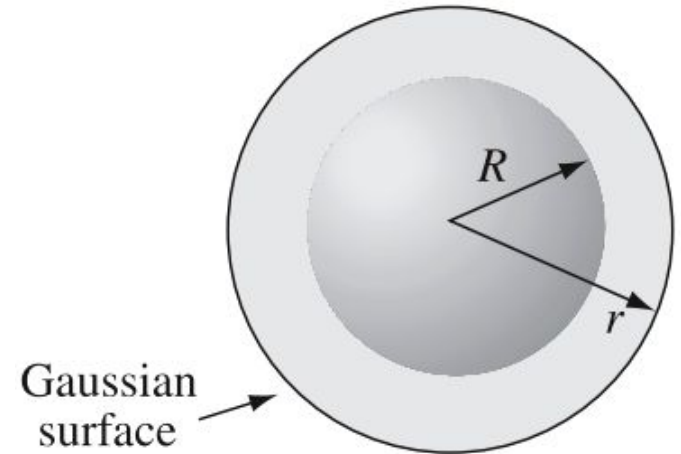
Solution:

Imagine a spherical surface at radius $r > R$.

This is called a **Gaussian surface** in the trade.

Gauss's law says that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$



and in this case, $Q_{\text{enc}} = q$.

At first glance, this doesn't seem to get us very far;

Because, the quantity we want ($\mathbf{E}$) is covered inside the surface integral.

Fortunately, symmetry allows us to extract $\mathbf{E}$ from under the integral sign:

$\mathbf{E}$ certainly points radially outward, as does $d\mathbf{a}$, so we can drop the dot product.

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da$$

and the *magnitude* of $\mathbf{E}$ is constant over the Gaussian surface, so it comes outside the integral:

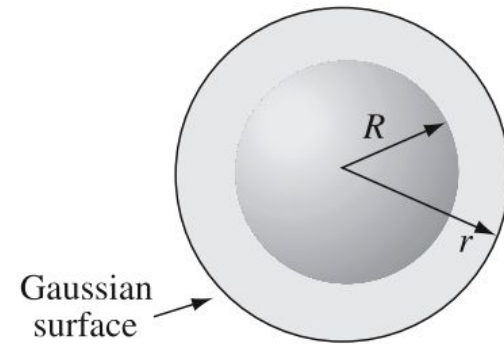
$$\int_S |\mathbf{E}| da = |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2.$$

Thus

$$|\mathbf{E}| 4\pi r^2 = \frac{1}{\epsilon_0} q,$$

or

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$



Notice a remarkable feature of this result:

The *field outside the sphere is exactly the same*, if all the *charge had been concentrated at the center*.

Gauss's law

Applications of Gauss's Law

Problem: An infinite plane carries a uniform surface charge σ . Find its **electric field**?

Solution: Draw a “Gaussian pillbox”, extending equal distances above and below the plane.

The surface with an infinite charge of uniform charge density used to determine the EF

Apply Gauss's law to this surface: $\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$

In this case, $Q_{\text{enc}} = \sigma A$

where, A is the area of the lid of the pillbox

By symmetry, $\mathbf{E}$ points away from the plane (upward for points above, downward for points below). So the top and bottom surfaces yield:

$$\int \mathbf{E} \cdot d\mathbf{a} = 2A|\mathbf{E}|$$

whereas the sides contribute nothing.

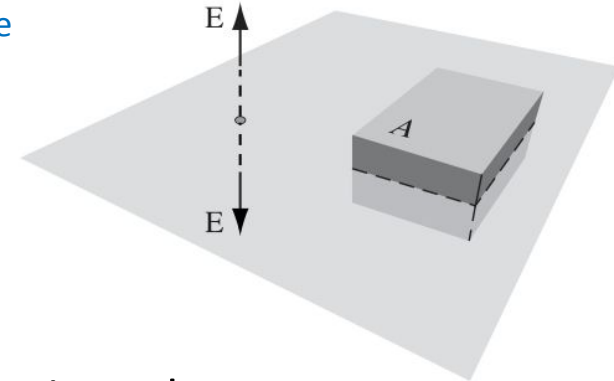
Thus

$$2A|\mathbf{E}| = \frac{1}{\epsilon_0} \sigma A$$

or,

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}}$ is a unit vector pointing away from the surface.

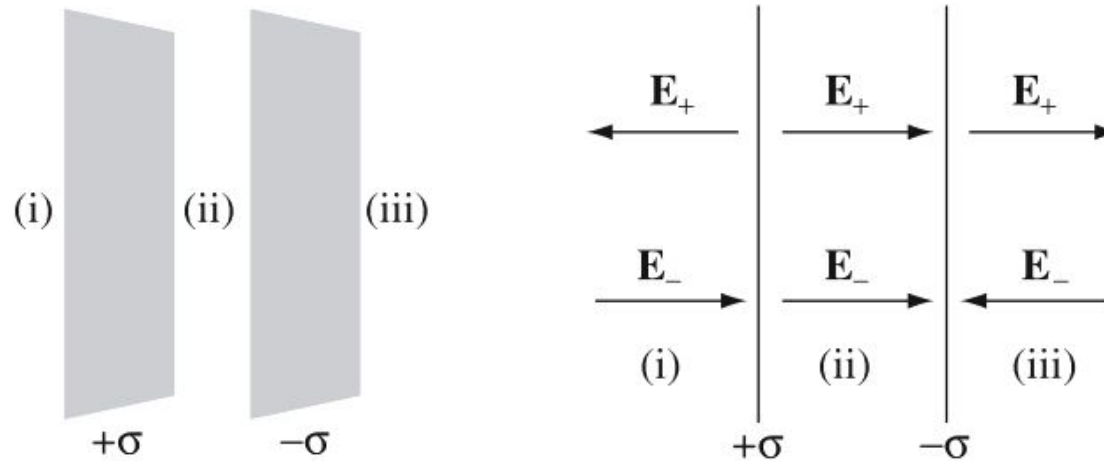


Note: The **field of an infinite plane** is **independent** of how far away you are.

Gauss's law

Applications of Gauss's Law

Problem: Two infinite parallel planes carry equal but opposite uniform charge densities, σ . Find the field in each of the three regions: (i) to the left of both, (ii) between them, (iii) to the right of both?

**Solution:**

The left plate produces a field $(1/2\epsilon_0)\sigma$, which points away from it to the left in region (i) and to the right in regions (ii) and (iii).

The right plate, being negatively charged, produces a field $(1/2\epsilon_0)\sigma$, which points toward it to the right in regions (i) and (ii) and to the left in region (iii).

The two fields cancel in regions (i) and (iii) and they conspire in region (ii).

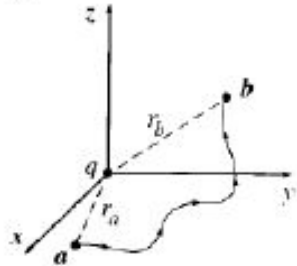
Conclusion: The field between the plates is σ/ϵ_0 , and points to the right; elsewhere it is zero.

The Curl of E

Consider the electric field from a point charge q at the origin: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$

Now let's calculate the line Integral of this field from some point $\mathbf{a}$ to some other point $\mathbf{b}$:

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}$$



In spherical coordinates, $d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin\theta d\phi \hat{\boldsymbol{\phi}}$

$$\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr \longrightarrow \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_{\mathbf{a}}^{\mathbf{b}} \frac{q}{r^2} dr = \left. \frac{-1}{4\pi\epsilon_0} \frac{q}{r} \right|_{r_a}^{r_b} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right)$$

The integral around a *closed* path is evidently zero (for then $r_a = r_b$):

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0 \rightarrow \text{Applying Stokes' theorem, } \nabla \times \mathbf{E} = 0$$

If we have many charges, the principle of superposition states that the total field is a vector sum of their individual fields: $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \dots$,

$$\nabla \times \mathbf{E} = \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \dots) = (\nabla \times \mathbf{E}_1) + (\nabla \times \mathbf{E}_2) + \dots = 0$$

$$\nabla \times \mathbf{E} = 0 \rightarrow \text{For any static charge distribution whatever}$$

Electric potential

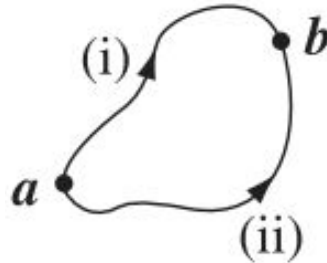
- The amount of work/energy needed to move electric charge from a reference point to the specific point in an electric field. $[V = k(q/r)]$
- We're going to exploit this special property of electric fields to reduce a vector problem (finding $\mathbf{E}$) to a much simpler scalar problem.

□ $\nabla \times \mathbf{E} = 0$, then the line integral of $\mathbf{E}$ around any closed loop is zero (that follows from Stokes' theorem).

$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

$$\oint_p \mathbf{E} \cdot d\mathbf{l} = \int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{a}$$

□ $\oint \mathbf{E} \cdot d\mathbf{l} = 0$, the line integral of $\mathbf{E}$ from point $\mathbf{a}$ to point $\mathbf{b}$ is the same for all paths.
(otherwise you could go out along path (i) & return along path (ii) to obtain)



□ The line integral is independent of path, we can define a function:

$$V(r) \equiv - \int_0^r \mathbf{E} \cdot d\mathbf{l}$$

○ Negative: same charges repel, work has to be done **against the Coulombic forces**

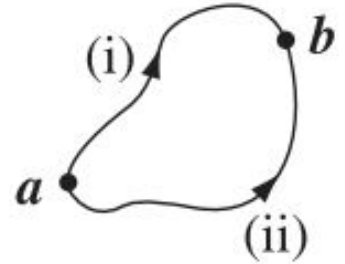
Here, $\mathbf{O}$ is some standard reference point on which we have agreed beforehand.

Then, V depends only on the point $\mathbf{r}$, it is called the electric potential.

Electric potential

□ The **potential difference** between **a** and **b** points is

$$\begin{aligned} V(\mathbf{b}) - V(\mathbf{a}) &= - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E} \cdot d\mathbf{l} \\ &= - \int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E} \cdot d\mathbf{l} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} \end{aligned}$$



□ The **fundamental theorem for gradient** states that

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

$$V(\mathbf{b}) - V(\mathbf{a}) = \int_a^b (\nabla V) \cdot d\mathbf{l}$$

or

$$\int_a^b (\nabla V) \cdot d\mathbf{l} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

- Electric potential is the **work done** to bring a unit charge from infinity to a given distance from **another unit positive charge**.
- Since like charges repel, work has to be done **against the Columbic forces** b/w the charges to bring them closer.
- **Work done against potential**. Hence, the **sign is opposite**.

✓ Finally, this is true for any points **a** and **b**, the integrands must be equal:

$$\mathbf{E} = -\nabla V$$

▪ A surface over which the potential is constant is called an **equipotential**.

Electric potential

Advantage of the potential formulation:

- If you know V , you can easily get $\mathbf{E}$, by just taking the gradient: $\mathbf{E} = -\nabla V$
- This feature has maximum advantage, **reducing a vector problem to a scalar one**, in which there is **no need to bother with the components**.

Potential obeys the superposition principle:

- The **original superposition principle** pertains to the force ($\mathbf{F}$) on a test charge Q .
- It says that the total force ($\mathbf{F}$) on Q is the vector sum of the forces attributable to the source charges individually. i.e. $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots$
- Dividing through by Q , we see that **the electric field, too, obeys the superposition principle**: $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \dots$ ($\mathbf{E} = \mathbf{F}/Q$)
- Integrating from the common reference point to r , **it follows that the potential also satisfies such a principle**: $V = V_1 + V_2 + \dots$ $V(r) \equiv -\int_o^r \mathbf{E} \cdot d\mathbf{l}$
- Hence, the **potential at any given point** is the **sum of the potentials due to all the source charges separately**.

✓ **Potential** has **units Nm/C**, or **J/C** (*Joule per coulomb*) and hence, **J/C** is a **volt**.

Electric potential

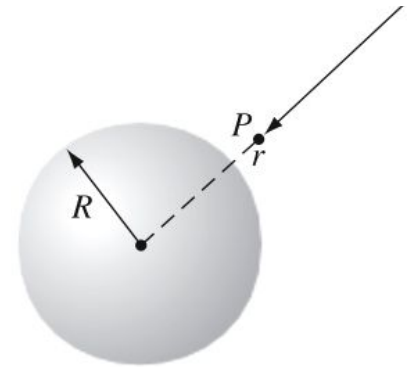
Problem: Find the **potential inside** and **outside** a **spherical shell** of radius R that carries a uniform surface charge. *Set the reference point at infinity.*

Solution:

From Gauss's law, the field outside is

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$



where q is the total charge on the sphere. The field inside is zero. For points outside the sphere ($r > R$),

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

To find the potential inside the sphere ($r < R$), we must break the integral into two pieces, using in each region the field that prevails there:

$$V(r) = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^R \frac{q}{r'^2} dr' - \int_R^r (0) dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^R + 0 = \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

Electric potential

Problem: An uniform electric field intensity is given by, $E = 15\hat{x}\text{kV/m}$. If the potential at the origin is zero, **calculate the potential at the point P(1,2,0)**

Solution:

$$V(r) = - \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$$

Given, $E = 15\hat{x}\text{kV/m}$

Also, $V=0$ at origin, then $x=0$

$$\begin{aligned} \text{Therefore, } V &= - \int_0^1 E \cdot d\mathbf{l} & d\vec{l} &= dx\hat{x} + dy\hat{y} + dz\hat{z} \Rightarrow d\vec{l} = dx\hat{x} \\ &= - \int_0^1 15 dx \\ &= -15\text{kV} \end{aligned}$$

Hence, potential V at P (1,2,0) = -15 kV

$$= -15 \times 10^3 \text{ V}$$

Electric potential

Electric potential of a localized charge distribution:

□ The electric field: $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$ $d\vec{l} = dr\hat{a}_r + r d\theta\hat{a}_\theta + r \sin\theta d\phi\hat{a}_\phi$

$d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin\theta d\phi \hat{\boldsymbol{\phi}}$ so, $\mathbf{E} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dr$ $V(r) \equiv - \int_O^r \mathbf{E} \cdot d\mathbf{l}$

- Setting the reference point (O) at infinity (∞), the potential of a point charge 'q' at the origin:

$$V(r) = - \int_O^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0} \int_\infty^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_\infty^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

- It is useful to remember that regions of positive charge are potential “hills”, regions of negative charge are potential “valleys”, and the electric field points “downhill” from plus to minus.

□ In general, the potential of a point charge q is $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

where r is the distance from q to $\mathbf{r}$.

- If we apply the superposition principle, then, the potential of a collection of charges:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i$

Electric potential Potential of a localized charge distribution

- For a **continuous distribution**:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r} dq$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

- In particular, **for a volume charge**:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$$

$$dq \rightarrow \lambda dl' \sim \sigma da' \sim \rho d\tau'$$

This equation tells us **how to compute V** as **we know ρ**, for a localized charge distribution.

- Similarly, the **potentials of line** and **surface charges** are

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda(\mathbf{r}')}{r} dl'$$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma(\mathbf{r}')}{r} da'$$

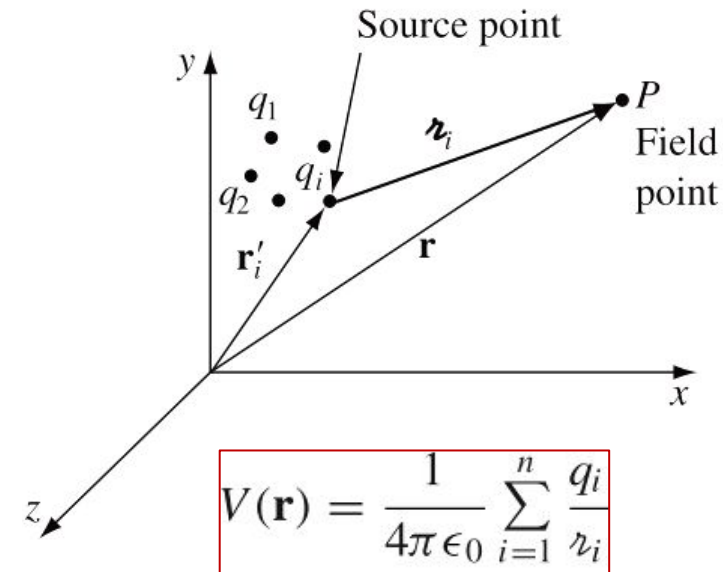
Electric potential

Electric potential due to discrete distribution of charges:

- Suppose, we have $q_1, q_2, \dots, q_n$ of the electric charges distributed discretely and $r_1, r_2, \dots, r_n$ are the position vectors, respectively.
- Then, the **electric potential** at point 'P' due to electric charge q_n is calculated as,

$$V_n(r) = \frac{1}{4\pi\epsilon_0} \frac{q_n}{(r - r_n)}$$

$$\mathbf{E}_n = \frac{1}{4\pi\epsilon_0} \frac{q_n}{r_n^2} \hat{\mathbf{r}}$$



$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

- The **total electrical potential** at point P is the **sum of all electrical potential** of all charges of distribution, $V = V_1 + V_2 + \dots + V_n$

$$V(r) = \sum_{i=1}^n \frac{K q_i}{(r - r_i)}$$

$$\text{Here, } K = \frac{1}{4\pi\epsilon_0}$$

Electric potential

Gradient of an electrostatic potential:

- We have two methods of determining potential, one directly from the **electric field intensity** by means of a **line integral**. $V(r) \equiv - \int_0^r E \cdot dl$ $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{r} d\tau'$
- Another from the **basic charge distribution** itself by a **volume integral**.
- Both methods are **helpful** in **determining the fields in most practical problems**.
- Perhaps, we wish to find the **capacitance between the conductors**, or the **charge and current distribution on the conductors** from which **losses may be calculated**.
- These quantities may be easily obtained **from the potential field** and **simple method of finding the electric field intensity** from the potential.
- We already have the **general line-integral relationship** between these quantities:

$$V = - \int \mathbf{E} \cdot d\mathbf{L}$$

$$V(\mathbf{r}) = - \int_0^r \mathbf{E} \cdot d\mathbf{l}$$

- The **operation on V by which $-\mathbf{E}$ is obtained** is known as the **gradient**, and the gradient of a scalar field T is defined as

$$\text{Gradient of } T = \text{grad } T = \frac{dT}{dN} \mathbf{a}_N$$

here, $\mathbf{a}_N$ is a unit vector normal to the **equipotential surfaces**.

Electric potential Gradient of an electrostatic potential

- Using this new term, now we may write the relationship between V and $\mathbf{E}$ as:

$$\mathbf{E} = -\text{grad } V$$

$$\mathbf{E} = -\nabla V$$

- Because, we have shown that V is a unique function of x , y , and z , we may take its total differential

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz$$

- But, we also have

$$dV = -\mathbf{E} \cdot d\mathbf{L} = -E_x dx - E_y dy - E_z dz$$

- Since the both expressions are true for any dx , dy , and dz , then:

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

- These results may be combined vectorially to yield:

$$\mathbf{E} = -\left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \right)$$

Electric potential

Gradient of an electrostatic potential

□ Comparing $\mathbf{E} = -\text{grad } V$ and $\mathbf{E} = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right)$ equations provides us with an expression which may be used **to evaluate** the **gradient in rectangular**

coordinates:

$$\text{grad } V = \frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Here,

$$\nabla = \frac{\partial}{\partial x}\mathbf{a}_x + \frac{\partial}{\partial y}\mathbf{a}_y + \frac{\partial}{\partial z}\mathbf{a}_z$$

□ The vector operator $\nabla T = \frac{\partial T}{\partial x}\mathbf{a}_x + \frac{\partial T}{\partial y}\mathbf{a}_y + \frac{\partial T}{\partial z}\mathbf{a}_z$ may be used formally as an operator.

□ From which we see that on a scalar, T , ∇T , producing $\nabla T = \text{grad } T$

□ This allows us to use a very compact expression to relate $\mathbf{E}$ and V :

$$\boxed{\mathbf{E} = -\nabla V}$$

- The **gradient** may be expressed in terms of **partial derivatives** in other coordinate systems :

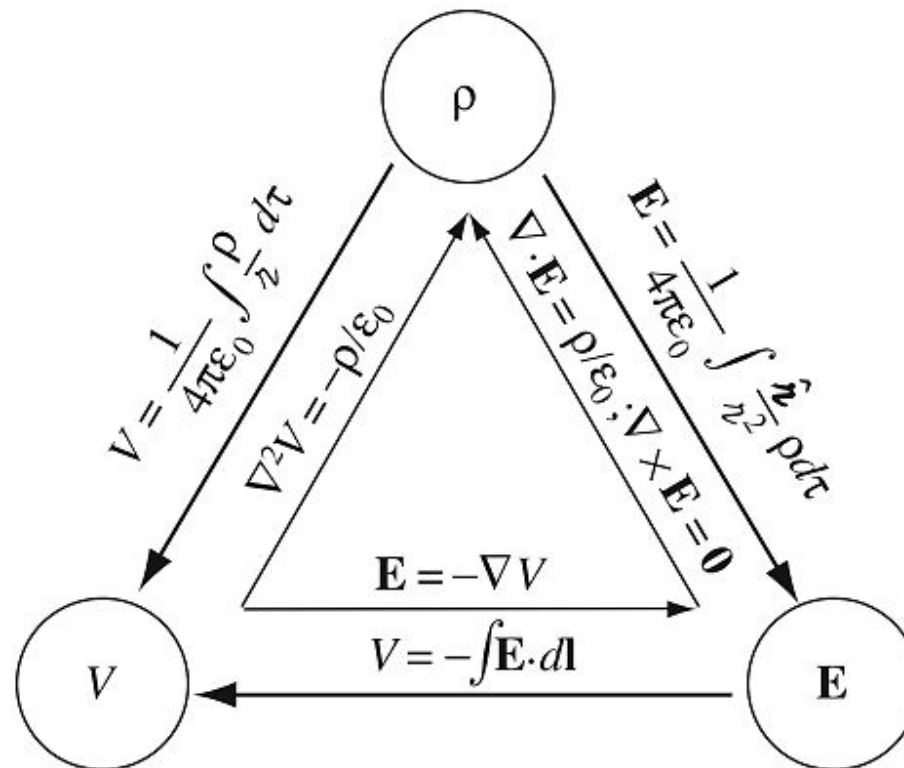
$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z \quad (\text{rectangular})$$

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z \quad (\text{cylindrical})$$

$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \quad (\text{spherical})$$

Boundary conditions

- In the **typical electrostatic problem**, from a given **source charge distribution ρ** , need to find the produced **electric field E** .
- Here, it is generally an **advantage to calculate the potential V** first, as an intermediate step.
- ✓ There are the **three fundamental quantities** of electrostatics: ρ , E , and V .
- ✓ If we want to derive all **six formulas** interrelating them, in the below figure are summarized.



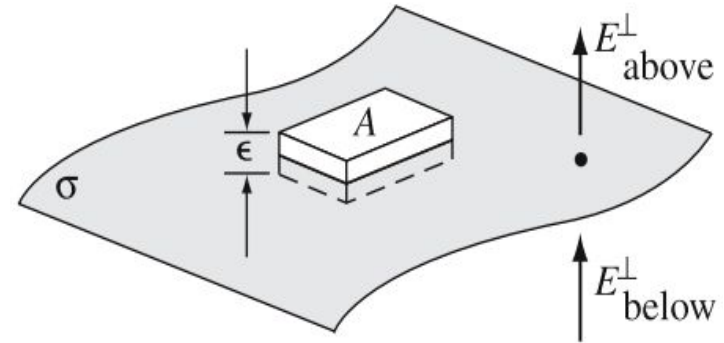
Boundary conditions

- We know that the **electric field**, $\mathbf{E}$ always undergoes a **discontinuity** when you cross a surface charge σ .
- It is a simple matter to find the amount by which $\mathbf{E}$ changes at **such a boundary**.
- Suppose, we draw a wafer-thin **Gaussian pillbox**, extending just barely over the edge in each direction.
The surface with an infinite charge of uniform charge density used to determine the electric field

Then, Gauss's law says that

$$\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \sigma A$$

where, A is the area of the pillbox-lid



- The **sides of the pillbox contribute nothing to the flux**, in the limit as the **thickness ϵ** goes to zero (*then A is negligible or ~ 1*), so we are left with

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma$$

Where, $E_{\text{above}}^{\perp}$ denotes the component of $\mathbf{E}$ that is perpendicular to the surface immediately above

and $E_{\text{below}}^{\perp}$ is the same, only just below the surface

Boundary conditions

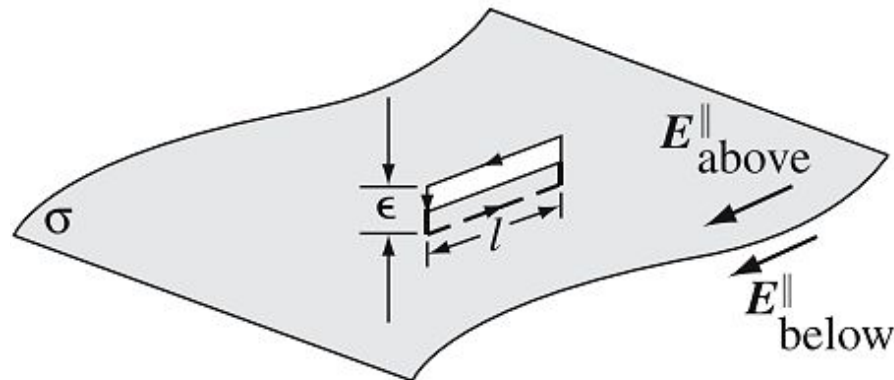
Conclusion:

- The normal component of $\mathbf{E}$ is discontinuous by an amount σ/ϵ_0 at any boundary.
- In particular, where there is no surface charge, $\mathbf{E}^\perp$ is continuous, as for instance at the surface of a uniformly charged solid sphere.
- The tangential component of $\mathbf{E}$, by contrast, is always continuous.

If we apply
$$\oint \mathbf{E} \cdot d\mathbf{l} = 0$$

Near the thin rectangular loop, the ends give nothing (as $\epsilon \rightarrow 0$) and the sides give

$(E_{\text{above}}^\parallel l - E_{\text{below}}^\parallel l)$ so, $\mathbf{E}_{\text{above}}^\parallel = \mathbf{E}_{\text{below}}^\parallel$ Where, $\mathbf{E}^\parallel$ stands for the components of $\mathbf{E}$ parallel to the surface.



Boundary conditions

- The **boundary conditions** on $\mathbf{E}$ can be combined into a single formula.

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

$$E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp} = \frac{1}{\epsilon_0} \sigma$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing from “below” to “above.”

- Similarly, the **potential** is **continuous across any boundary**

since,

$$V_{\text{above}} - V_{\text{below}} = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l}$$

$$V(r) \equiv - \int_0^r E \cdot dl$$

- As the **path length shrinks to zero**, so too does the integral:

$$V_{\text{above}} = V_{\text{below}}$$

- However, **the gradient of V** inherits the **discontinuity in $\mathbf{E}$** .

- Since, $\mathbf{E} = -\nabla V$, implies that

$$\nabla V_{\text{above}} - \nabla V_{\text{below}} = -\frac{1}{\epsilon_0} \sigma \hat{\mathbf{n}}$$

Boundary conditions

□ In more conveniently,

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0}\sigma$$

$$\nabla V_{\text{above}} - \nabla V_{\text{below}} = -\frac{1}{\epsilon_0}\sigma \hat{\mathbf{n}}$$

where,

$$\frac{\partial V}{\partial n} = \nabla V \cdot \hat{\mathbf{n}}$$

It denotes the **normal derivative** of V

i.e., **the rate of change in the direction perpendicular to the surface.**

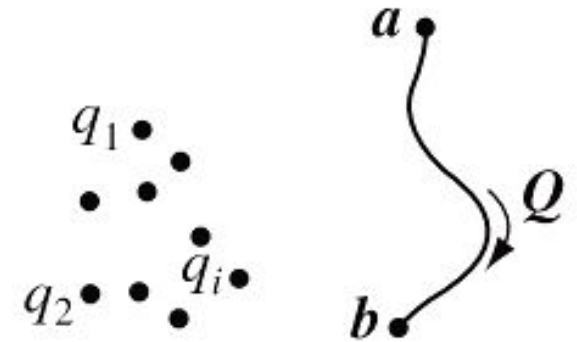
❖ Note that these boundary conditions relate the **fields** and **potentials** *just above* and *just below the surface*.

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{1}{\epsilon_0}\sigma$$

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

Energy for a charge distribution

- Suppose you have a stationary configuration of source charges, and you want to move a test charge Q from point 'a' to point 'b'.



Question: How much work will you have to do?

- At any point along the path, the electric force on Q is $\mathbf{F} = Q\mathbf{E}$. ($\mathbf{E} = \mathbf{F}/Q$)
- The force you must exert, in opposition to this electrical force, is $-\mathbf{Q}\mathbf{E}$.

❖ Therefore, the required work is ($W = \mathbf{f} \cdot \mathbf{d}$)

$$V_{\text{above}} - V_{\text{below}} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

$$W = \int_a^b \mathbf{F} \cdot d\mathbf{l} = -Q \int_a^b \mathbf{E} \cdot d\mathbf{l} = Q[V(\mathbf{b}) - V(\mathbf{a})]$$

- Notice that the *answer is independent of the path* that take from $\mathbf{a}$ to $\mathbf{b}$.
- In mechanics, we may call as the electrostatic force “conservative”.
- Dividing through by Q , we have

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q}$$

Energy for a charge distribution

- Hence, the **potential difference** between points **a** and **b** is equal to the **work per unit charge** required to carry a particle from **a** to **b**.

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q}$$

- In particular, if you want to bring Q in from far away (*say infinite*) and stick it at point ' $\mathbf{r}$ ', the work you must do is

$$W = Q[V(\mathbf{r}) - V(\infty)]$$

- If you have set the reference point at infinity, then $W = QV(\mathbf{r})$.
- In this sense, **potential is potential energy per unit charge**.

The Energy of a Point Charge Distribution:

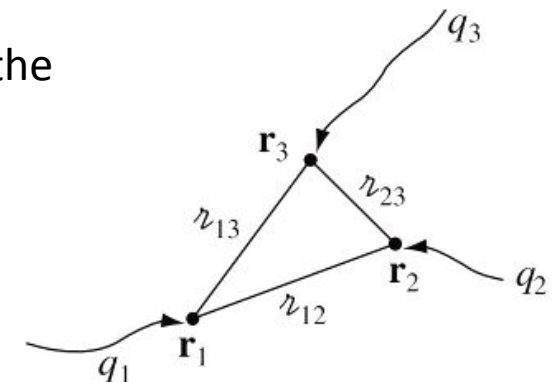
How much **work** would it take to assemble an entire collection of point charges?

- Imagine, the charges can **bring one by one from far away**.
- The first charge, q_1 , **takes no work**, since there is **no field** yet to fight against.

- Now bring in q_2 , this will cost you $q_2 V_1(\mathbf{r}_2)$, where V_1 is the potential due to q_1 , and $\mathbf{r}_2$ is the place we're putting q_2 .

$$W_2 = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right)$$

r_{12} is the distance between q_1 and q_2 once they are in position.



Energy of a point charge distribution

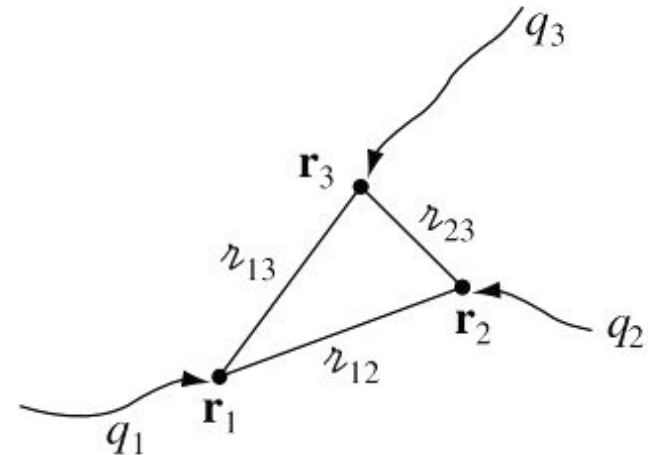
□ Now bring in q_3 ; this requires work $q_3 V_{1,2}(r_3)$, where $V_{1,2}$ is the potential due to charges q_1 and q_2 , namely, $(1/4\pi\epsilon_0)(q_1/r_{13} + q_2/r_{23})$

So

$$W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right)$$

□ Similarly, the extra work to bring in q_4 will be

$$W_4 = \frac{1}{4\pi\epsilon_0} q_4 \left(\frac{q_1}{r_{14}} + \frac{q_2}{r_{24}} + \frac{q_3}{r_{34}} \right)$$



□ The **total work** necessary to collect the first four charges, then

$$W = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right)$$

□ Now, take the product of each pair of charges, divide by their separation distance, and add it all up:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}$$

Energy of a point charge distribution

- The condition $j > i$ is to remind you **not to count the same pair twice**.
- Simply, to accomplish this, intentionally **to count each pair twice**, and then **divide by 2**:

$$W = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}$$

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}$$

Here, we must still avoid $i = j$

- Finally, let's pull out the factor q_i :

$$W = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j \neq i}^n \frac{1}{4\pi\epsilon_0 r_{ij}} q_j \right)$$

- The term in parentheses is the **potential at point $\mathbf{r}_i$** (the position of q_i) **due to all the other charges**.

Thus,

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$$

Energy of a continuous charge distribution

- For a volume charge density ρ ,

$$W = \frac{1}{2} \int \rho V d\tau$$

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$$

- The integrals for line & surface charges can be $\int \lambda V dl$ and $\int \sigma V da$
- There is an easy way to rewrite this result, in which ρ and V are eliminated in favor of $\mathbf{E}$.
- First, use Gauss's law to express ρ in terms of $\mathbf{E}$:

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$$

$$\rho = \epsilon_0 \nabla \cdot \mathbf{E}, \quad \text{so} \quad W = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau$$

- Now, use integration by parts to transfer the derivative from $\mathbf{E}$ to V :

$$W = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) d\tau + \oint V \mathbf{E} \cdot d\mathbf{a} \right]$$

Energy of a continuous charge distribution

- We know that $\nabla V = -\mathbf{E}$.

$$W = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) d\tau + \oint V \mathbf{E} \cdot d\mathbf{a} \right]$$

Hence,
$$W = \frac{\epsilon_0}{2} \left(\int_V E^2 d\tau + \oint_S V \mathbf{E} \cdot d\mathbf{a} \right)$$

□ If we integrate over all space, then the **surface integral goes to zero**, then

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

Problem: Find the **energy of a uniformly charged spherical shell** of total charge q and radius R ?

Solution: Inside the sphere, $\mathbf{E} = 0$

Outside,
$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}, \quad \text{so} \quad E^2 = \frac{q^2}{(4\pi\epsilon_0)^2 r^4}$$

$$\begin{aligned} W_{\text{tot}} &= \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int_{\text{outside}} \left(\frac{q^2}{r^4} \right) (r^2 \sin\theta dr d\theta d\phi) \\ &= \frac{1}{32\pi^2\epsilon_0} q^2 4\pi \int_R^\infty \frac{1}{r^2} dr = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}. \end{aligned}$$

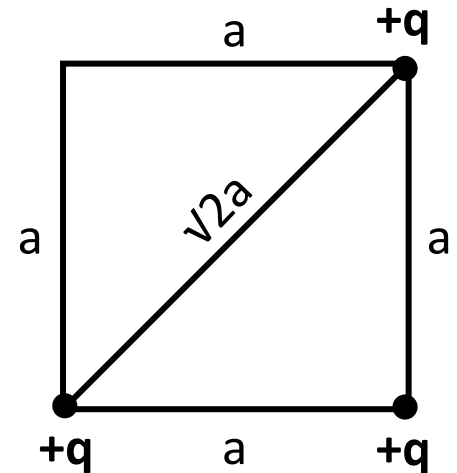
Energy for a charge distribution

Problem: How much work does it take to bring another charge $+q$ from far away to place it in the fourth corner, if three charges $(+q)$ are situated at the corners of a square of side 'a'?

Solution: For initial configurations, the potential energy of the system:

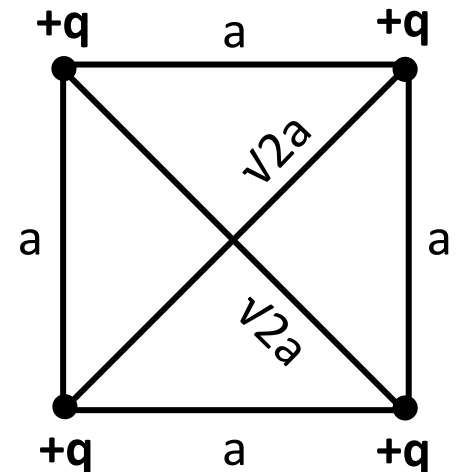
$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}$$

$$\begin{aligned} W_1 &= \frac{1}{4\pi\epsilon_0} \left(\frac{q \cdot q}{a} + \frac{q \cdot q}{a} + \frac{q \cdot q}{\sqrt{2}a} \right) \\ &= \frac{q^2}{4\pi\epsilon_0 a} \left(2 + \frac{1}{\sqrt{2}} \right) \end{aligned}$$



For final configurations, the potential energy of the system:

$$\begin{aligned} W_2 &= \frac{1}{4\pi\epsilon_0} \left(\frac{q \cdot q}{a} + \frac{q \cdot q}{a} + \frac{q \cdot q}{a} + \frac{q \cdot q}{a} + \frac{q \cdot q}{\sqrt{2}a} + \frac{q \cdot q}{\sqrt{2}a} \right) \\ &= \frac{q^2}{4\pi\epsilon_0 a} \left(4 + \frac{2}{\sqrt{2}} \right) \end{aligned}$$



We know, work done is the change in potential energy.

Therefore,
$$W = W_2 - W_1 = \frac{q^2}{4\pi\epsilon_0 a} \left(4 + \frac{2}{\sqrt{2}} - 2 - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{q^2}{4\pi\epsilon_0 a} \left(2 + \frac{1}{\sqrt{2}} \right)$$

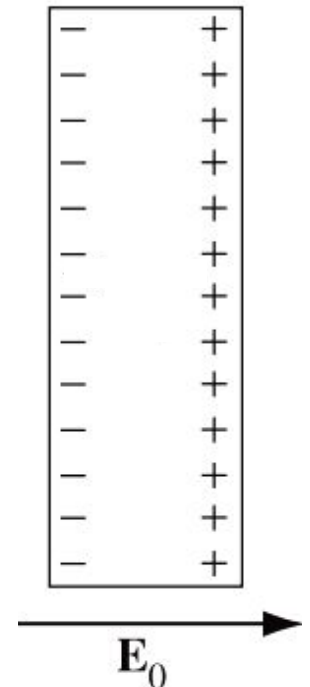
Conductors

- In an **insulator**, such as glass or rubber, each electron is on a short direction, **attached to a particular atom** (**tightly bound**).
- In a **metallic conductor**, one or **more electrons per atom** are **free to move**.
- A perfect conductor would contain an **unlimited supply of free charges**.
- In reality, **there are no perfect conductors**, but **metals come pretty close**, for most purposes.

The basic electrostatic properties of ideal conductors:

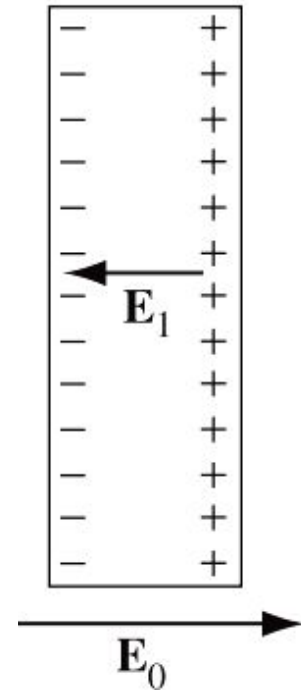
(i) $E = 0$ inside a conductor:

- If there is any field, **those free charges would move**, and **it wouldn't be electrostatics** any more.
- **What happens when you put a conductor into an external electric field E_0 ?**
- Initially, the field will drive any **free positive charges** to the **right** and **negative ones** to the left.



Conductors

- When they come to the edge of the material, the charges pile up: Plus on the right side and minus on the left.
- Now, these induced charges produce a field of their own, E_1 , which (see figure) is in the opposite direction to E_0 .
- It means that the field of the induced charges tends to cancel the original field.
- Charge will continue to flow until this cancellation is complete, and the resultant field inside the conductor is precisely zero.
- The whole process is *practically instant/immediate*.



(ii) $\rho = 0$ inside a conductor: ρ : Electric charge per unit area of a surface, or
Electric charge per unit volume of a body or field

- This follows from Gauss's law: $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$
If E is zero, then ρ also is zero.
But, there is still charge around and it exactly as much plus as minus.
Hence, the net charge density in the interior is zero.

Conductors

(iii) A conductor is an equipotential:

If **a** and **b** are any two points within (or at the surface of) a given conductor,

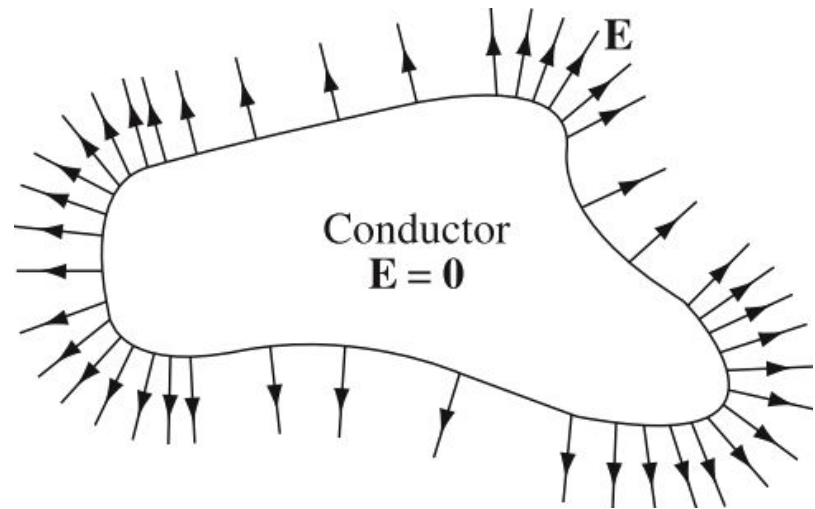
$$V(\mathbf{b}) - V(\mathbf{a}) = - \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = 0,$$

and hence $V(\mathbf{a}) = V(\mathbf{b})$

(iv) E is perpendicular to the surface, just outside a conductor:

Otherwise as in (i): $E=0$ inside a conductor, charge will immediately flow around the surface until it kills off the tangential component.

Perpendicular to the surface, charge cannot flow.

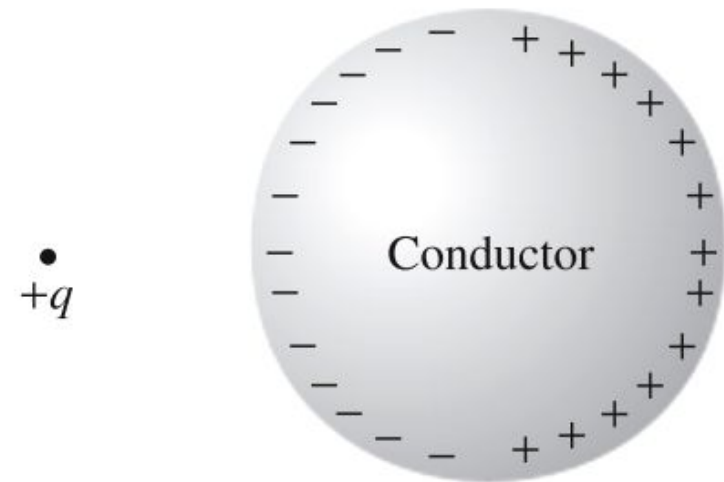


(V) Any net charge resides on the surface: That is the only place left/absent and will discuss now. (induced charges)

Conductors

Induced Charges:

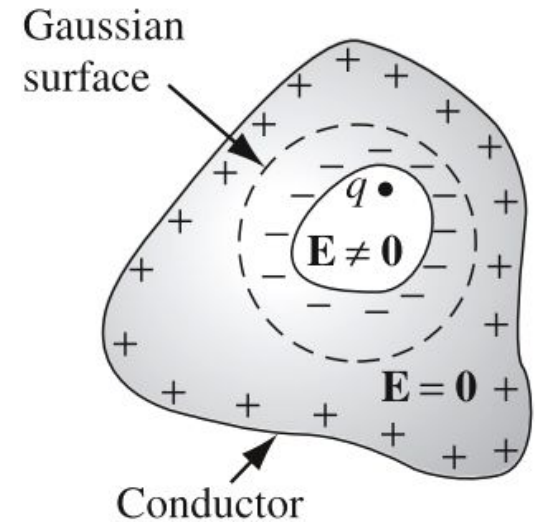
- If you hold a charge $+q$ near an uncharged conductor, the two will attract one another.
- The reason for this is that $+q$ will pull minus $(-)$ charges over to the near side and repel plus $(+)$ charges to the far side.
- Since, the negative induced charge is closer to $+q$, there is a net force of attraction.
- If there is some hollow cavity in the conductor, and within that cavity you put some charge, then the field in the cavity will not be zero.
- In a remarkable way, the cavity and its contents are electrically isolated from the outside world by the surrounding conductor.
- No external fields penetrate the conductor.
- They are canceled at the outer surface by the induced charge there.



Conductors

Induced Charges

- Similarly, the *field due to charges within the cavity* is *canceled*, by the induced charge on the **inner surface**.
- The **compensating charge left over** on the **outer surface** of the conductor effectively “**communicates**” the **presence of +q** to the outside world.
- The **total charge** induced on the cavity wall is **equal** and **opposite to charge inside**.



- If we **surround the cavity with a Gaussian surface**, **all points of which are in the conductor** $\oint \mathbf{E} \cdot d\mathbf{a} = 0$ and hence (by Gauss's law) the **net enclosed charge must be zero**.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

- But $Q_{\text{enc}} = q + q_{\text{induced}}$
- Hence, $q_{\text{induced}} = -q$

- Then if the **conductor as a whole** is **electrically neutral**, there must be a **charge +q** on its **outer surface**.

Conductors

Surface Charge and the Force on a Conductor:

- Because the field inside a conductor is zero, boundary condition requires that the field immediately outside is

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{\text{below}} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

Consistent with our earlier conclusion that the field is normal to the surface.

In terms of potential

$$\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$$

$$\mathbf{E} = -\nabla V$$

These equations help to calculate the surface charge on a conductor.

- In the presence of an electric field, a surface charge will experience a force, i.e., the force per unit area, $\mathbf{f} = \sigma \mathbf{E}$. ($F = qE$)
- But there's a problem here, for the electric field is discontinuous at a surface charge, so what are we supposed to use: $\mathbf{E}_{\text{above}}$, $\mathbf{E}_{\text{below}}$ or something in between?
- ✓ The answer is that we should use the average of the two:

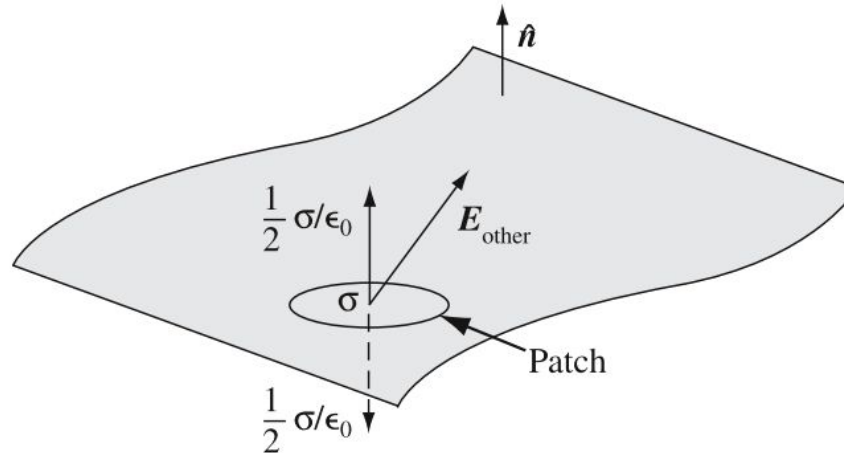
$$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2} \sigma (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}})$$

Surface Charge and the Force on a Conductor

Why the average?

$$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2} \sigma (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}})$$

✓ Let's focus our attention on a **tiny patch of surface** surrounding the point.



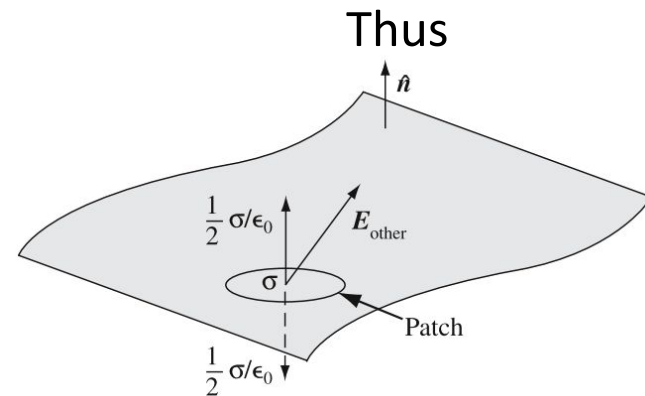
- The **total field consists of two parts**, that attributable to the **patch itself**, and that due to **everything else**: $\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}$
- The **patch cannot exert a force on itself**, any more than you can lift yourself by standing in a basket and pulling up on the handles.
- The force on the patch is **due to exclusively** $\mathbf{E}_{\text{other}}$.

Surface Charge and the Force on a Conductor

- ❖ The discontinuity is due **entirely to the charge on the patch**, which puts out a field on either side, **pointing away from the surface**.

$$\mathbf{f} = \sigma \mathbf{E}_{\text{average}} = \frac{1}{2} \sigma (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}})$$

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$



$$\mathbf{E}_{\text{above}} = \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}},$$

$$\mathbf{E}_{\text{below}} = \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}},$$

Then

$$\mathbf{E}_{\text{other}} = \frac{1}{2} (\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}) = \mathbf{E}_{\text{average}}$$

- In the particular case of a conductor, the **field is zero inside** and $(\sigma/\epsilon_0)\hat{\mathbf{n}}$ **outside** so the **average** is $(\sigma/2\epsilon_0)\hat{\mathbf{n}}$ and

the **force per unit area** is

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{\mathbf{n}}$$

- ✓ This amounts to an **outward electrostatic pressure on the surface**, tending to **draw the conductor into the field**, regardless of the sign of σ .
- ✓ Expressing the **pressure in terms of the field**, just outside the surface:

$$P = \frac{\epsilon_0}{2} E^2$$

Capacitors

✓ Suppose we have two conductors, and we put charge $+Q$ on one and $-Q$ on the other.

- Since V is constant over a conductor, then the potential difference between them is

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \mathbf{E} \cdot d\mathbf{l}$$



- We don't know how the charge distributes itself over the two conductors, and calculating the field would be tough.
- If their shapes are complicated, then $\mathbf{E}$ is proportional to Q .
- For $\mathbf{E}$ is given by Coulomb's law:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r^2} \hat{\mathbf{r}} d\tau$$

Here, if we want to double the ρ , we need to double $\mathbf{E}$

Capacitors

- Since **E** is proportional to **Q**, and also proportional to **V**.
- The constant of proportionality is called the capacitance:

$$C \equiv \frac{Q}{V}$$



- Capacitance is a purely geometrical quantity, determined by the sizes, shapes, and separation of the two conductors.
- In SI units, C is measured in farads (F) and Farad is a C/V.
- More practical units are the microfarad (10^{-6} F) and the picofarad (10^{-12} F).
- ✓ Potential **V** is the potential of the positive conductor less that of the negative one.
- ✓ Capacitance **C** is an intrinsically positive quantity.

Capacitors

- To “charge up” a capacitor, we have to remove electrons from the positive plate and carry them to the negative plate.
- In this case, it is against the electric field, which is pulling them back toward the positive conductor and pushing them away from the negative one.
- How much work does it take to charge the capacitor up to a final amount Q ?
- Suppose, at some intermediate stage in the process the charge on the positive plate is q , so that the potential difference is q/C . ($C=Q/V$, then $V=Q/C$)
- ✓ The work need to transport the next piece of charge, dq , is

$$dW = \left(\frac{q}{C}\right) dq$$

$$W = QV$$

Then the total work required to go from $q = 0$ to $q = Q$ is

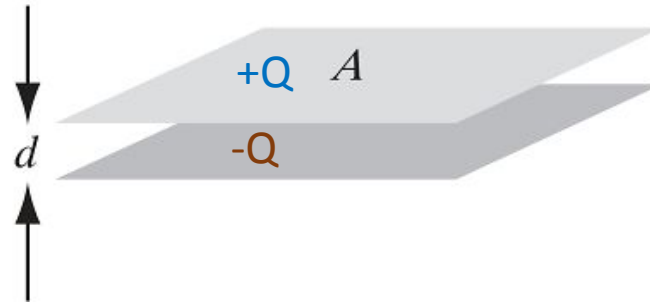
$$W = \int_0^Q \left(\frac{q}{C}\right) dq = \frac{1}{2} \frac{Q^2}{C}$$

$$W = \frac{1}{2} C V^2$$

since $Q = CV$, where V is the final potential of the capacitor

Capacitors

Problem: Find the **capacitance** of a parallel-plate capacitor consisting of two metal surfaces of area 'A' held a distance 'd' apart?



$$C \equiv \frac{Q}{V}$$

Solution:

If we put **+Q on the top** and **-Q on the bottom**, charges will spread out uniformly over the two surfaces, provided the area is reasonably large and the separation small.

Then the surface charge density is $\sigma = Q/A$ on the top plate

and the field is, $E = (1/\epsilon_0) Q/A$

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

The potential difference (V) between the plates is therefore, $V = \frac{Q}{A\epsilon_0} d$

Hence,

$$C = \frac{A\epsilon_0}{d}$$

$$C \equiv \frac{Q}{V}$$

Ex: The plates are square with **sides 1 cm long**, and they are **held 1 mm apart**

Then the **capacitance** is **$9 \times 10^{-13} \text{ F}$**

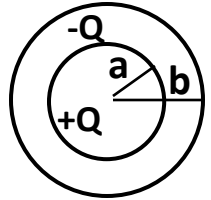
Capacitors

Problem: Find the **capacitance** of two concentric spherical metal shells, with radii 'a' and 'b'?

Solution: Place charge **+Q** on the inner sphere, and **-Q** on the outer one.

The field between the spheres is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$$



Hence, the potential difference between them is

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \frac{Q}{4\pi\epsilon_0} \int_b^a \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

V is proportional to Q, then the capacitance is

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{(b - a)}$$

Poisson's Equation & Laplace's Equation

Poisson's Equation & Laplace's Equation:

Poisson's Equation & Laplace's Equation:

□ Electric field can be written as the gradient of a scalar potential: $\mathbf{E} = -\nabla V$

□ Then the divergence and curl of $\mathbf{E}$ can be: $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ and $\nabla \times \mathbf{E} = \mathbf{0}$

- In terms of V , $\nabla \cdot \mathbf{E} = \nabla \cdot (-\nabla V) = -\nabla^2 V$.
- So, apart from that persistent minus sign, the divergence of $\mathbf{E}$ is the Laplacian of V .

- Then, Gauss's law says,

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

- This is known as Poisson's equation.
- In regions where there is no charge, so $\rho = 0$.
- Then Poisson's equation reduces to Laplace's equation $\nabla^2 V = 0$.

Rectangular:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Cylindrical:

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

Spherical:

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

Laplace's equation

□ Poisson's equation

$$\nabla^2 V = -\frac{1}{\epsilon_0} \rho$$

- We are interested in finding the **potential in a region**, where $\rho = 0$.
- In this case, **Poisson's equation reduces** to **Laplace's equation**: $\nabla^2 V = 0$
- This formula is so fundamental to the subject that one might almost say **electrostatics is the study** of **Laplace's equation**.

Uniqueness Theorem

Uniqueness Theorem: If a solution to Laplace's equation can be found that satisfies the boundary conditions, then the solution is unique.

Solving the Laplace and Poisson equations

The guaranteed uniqueness of solutions has spawned several creative ways to solve the Laplace and Poisson equations for the electric potential. We will treat three of them in this class:

- ❑ **Method of images** (today).

Very powerful technique for solving electrostatics problems involving charges and conductors.

- ❑ **Separation of variables**

Perhaps the most useful technique for solving partial differential equations. You'll be using it frequently in quantum mechanics too.

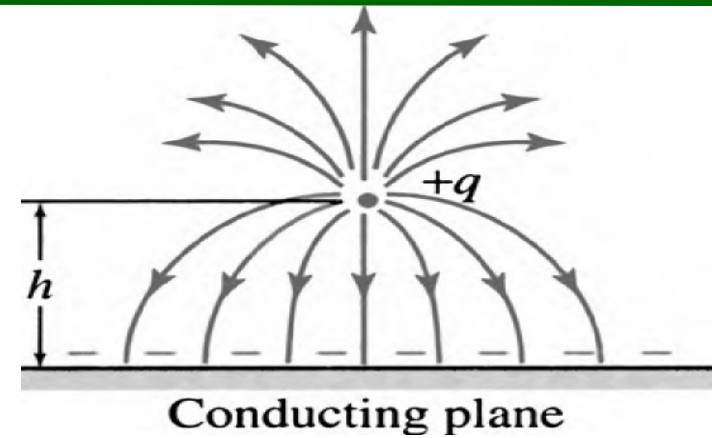
- ❑ **Multipole expansion**

Fermi used to say, "When in doubt, expand in a power series." This provides another fruitful way to approach problems not immediately accessible by other means.

The Image Charge

- Potential at any point on the plane bisecting the dipole is zero, and electric field intensity is normal to the plane.

- bisecting plane $\equiv$ a conducting plane.



- Field pattern of dipole remains unchanged, if a conducting plane in place of bisecting plane
- Field distribution does not change with removal of negative charge below the conducting plane
- Total charge induced on the surface of the conductor is $-q$,
- Reverse is also true.
- For two bisecting planes number of images will be finite as long as the angle between the planes is a submultiple of 360° .
- If ϑ is the angle of intersection of two planes, field due to a point charge placed between the planes can be obtained by replacing planes with $n - 1$ image charges as long as $n = 360^\circ/\vartheta$ is an integer.

Image problems -1

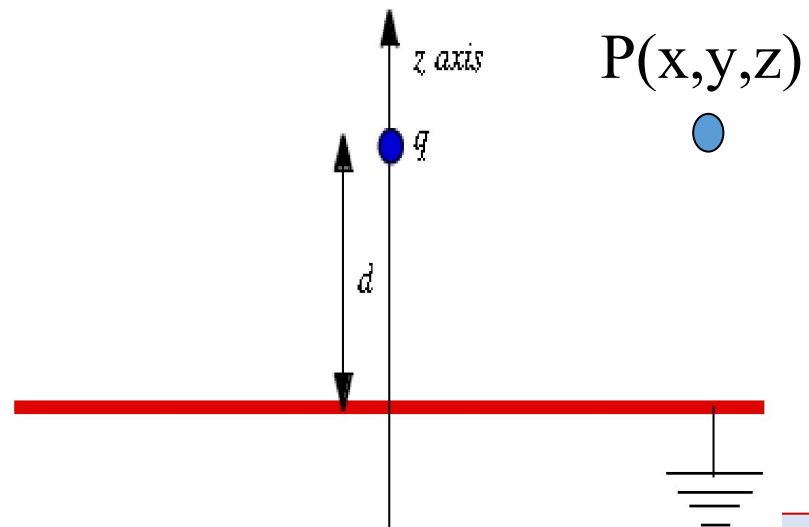
Consider a point charge q held as a disc above **an infinite grounded conducting**

The electrostatic potential of this system satisfy the following **two boundary con**

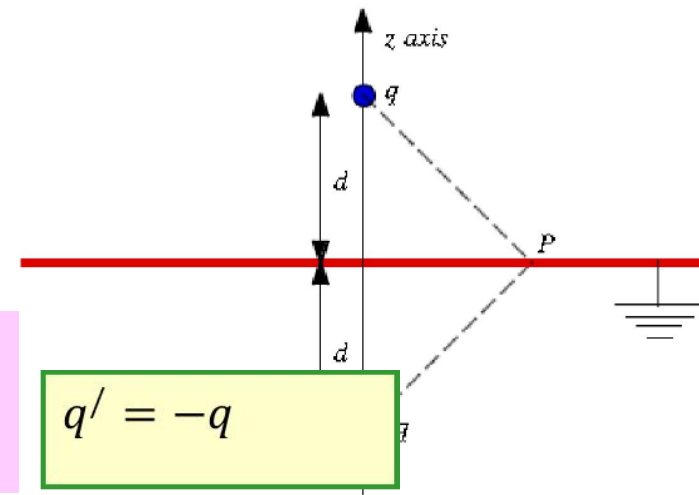
- 1 $V(x, y, 0) = 0$

- 2 $V(x, y, z) \rightarrow 0; \text{ as } x, y, z \rightarrow \infty$

- A direct calculation of the electrostatic potential and charge distribution on the grounded conductor is unknown



Consider a second system, consisting of two point charges with charges $+q$ and q' , located at $z = d$ and $z = -d$, respectively.



$$V(x, y, 0) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + d^2}} + \frac{q'}{\sqrt{x^2 + y^2 + d^2}} \right] = 0$$

Image problems - 1

The potential of this system at infinity will approach zero since the potential generated by each charge will decrease as $1/r$ with increasing distance r . Therefore, **the electrostatic potential generated by the two charges satisfies the same boundary conditions.**

$$\vec{E}_{outside} = \frac{\sigma}{\epsilon_0} \hat{n}$$

$$\sigma = \epsilon_0 E_z = -\epsilon_0 \left. \frac{\partial V}{\partial z} \right|_{z=0}$$

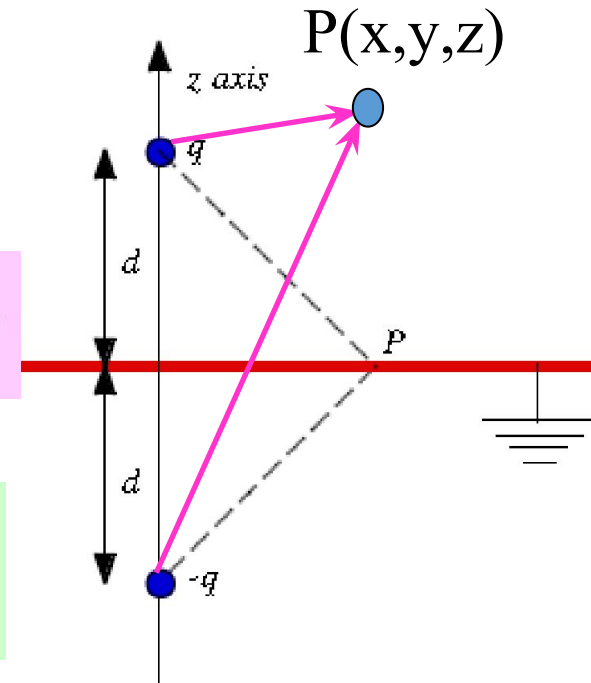
$$\sigma = -\frac{q}{4\pi} \left\{ \frac{-(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right\} \Big|_{z=0}$$

$$Q_{total} = \int_{Surface} \sigma da = \int_0^{2\pi} \int_0^{\infty} \sigma(r) r dr d\theta$$

$$Q_{total} = -qd \int_0^{\infty} \frac{1}{(r^2 + d^2)^{3/2}} r dr = \frac{qd}{\sqrt{r^2 + d^2}} \Big|_0^{\infty} = qd \left[0 - \frac{1}{d} \right] = -q$$

$$\sigma = -\frac{q}{2\pi} \frac{d}{(x^2 + y^2 + d^2)^{3/2}}$$

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} + \frac{-q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$

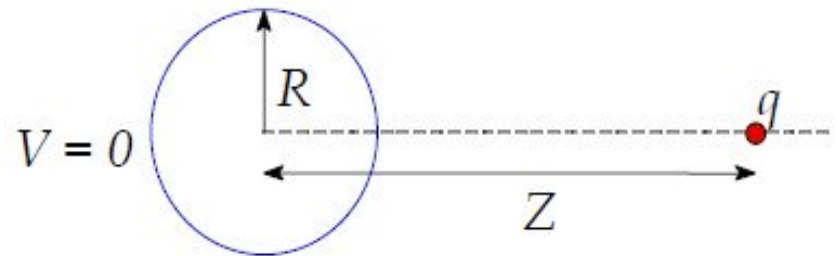


Example - 2

Potential for a point charge and a grounded sphere (Example 3.2 + Problem 3.7 in Griffiths)

A point charge q is situated a distance Z from the center of a grounded conducting sphere of radius R .

- ☐ Find the potential everywhere.
- ☐ Find the induced surface charge on the sphere, as function of θ . Integrate this to get the total induced charge.
- ☐ Calculate the potential energy of the system.



Example - 2

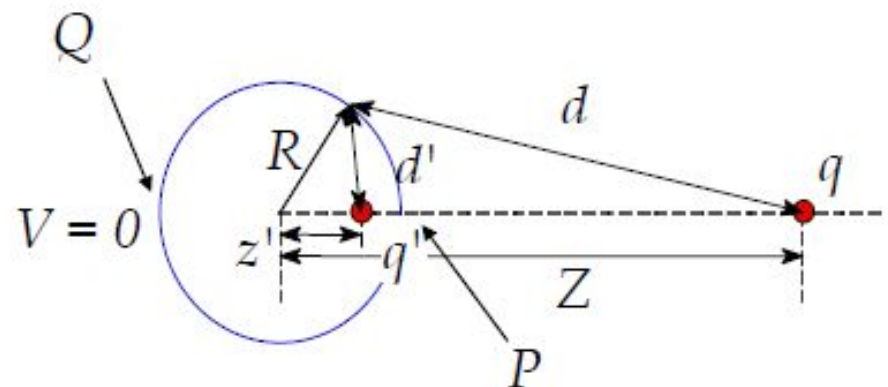
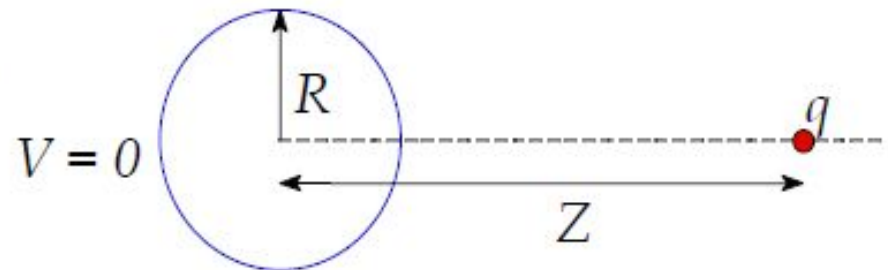
We need to find a position to put charge q' such that the boundary conditions on the sphere are satisfied. Start by determining where this charge should lie, using the points along the z axis:

$$V_P = 0 = \frac{q}{Z - R} + \frac{q'}{R - z'}$$

$$\Rightarrow q' = -\frac{q}{Z - R}(R - z')$$

$$V_Q = 0 = \frac{q}{Z + R} + \frac{q'}{R + z'}$$

$$\Rightarrow q' = -\frac{q}{Z + R}(R + z')$$



Example - 2

$$\text{So } \frac{q}{Z-R}(R-z') = \frac{q}{Z+R}(R+z')$$

$$(Z+R)(R-z') = (Z-R)(R+z')$$

Rearrange:

$$z'(Z-R) + z'(Z+R) = 2Zz'$$

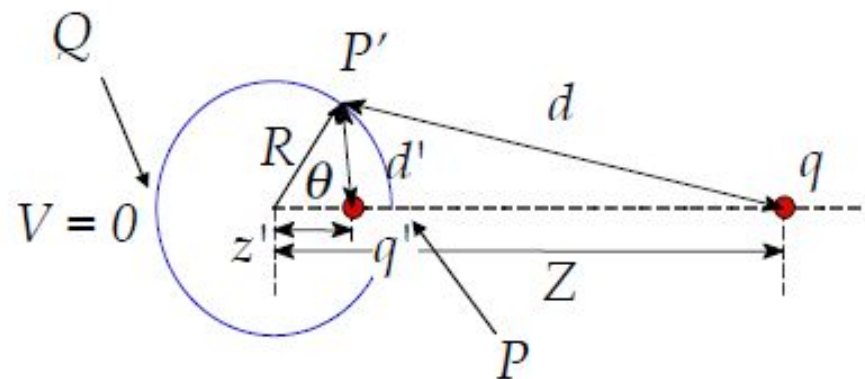
$$= R(Z+R) - R(Z-R) = 2R^2$$

$$\Rightarrow z' = \frac{R^2}{Z}, \quad q' = -\frac{q}{Z+R}(R+z') = -\frac{q}{Z+R}\left(R + \frac{R^2}{Z}\right) = -q \frac{R}{Z}$$

Now consider an arbitrary point on the sphere, P' :

$$d = \sqrt{R^2 + Z^2 - 2RZ \cos \theta}$$

$$d' = \sqrt{R^2 + z'^2 - 2Rz' \cos \theta} = \sqrt{R^2 + \frac{R^4}{Z^2} - 2R \frac{R^2}{Z} \cos \theta}$$



Example - 2

The potential should come out to be zero there, and sure enough,

$$\begin{aligned} V_{P'} &= \frac{q}{d} + \frac{q'}{d'} = \frac{q}{\sqrt{R^2 + Z^2 - 2RZ \cos \theta}} + \frac{-q \frac{R}{Z}}{\sqrt{R^2 + \frac{R^4}{Z^2} - 2R \frac{R^2}{Z} \cos \theta}} \\ &= \frac{q}{\sqrt{R^2 + Z^2 - 2RZ \cos \theta}} + \frac{-q \frac{R}{Z}}{\sqrt{R^2 + \frac{R^4}{Z^2} - 2R \frac{R^2}{Z} \cos \theta}} \\ &= \frac{q}{\sqrt{R^2 + Z^2 - 2RZ \cos \theta}} - \frac{q}{\sqrt{R^2 + Z^2 - 2RZ \cos \theta}} = 0 \end{aligned}$$

Thus the potential outside the grounded sphere is given by the superposition of the potential of the charge q and the image charge q' .

Example - 2

So the potential at some point (r, θ, ϕ) outside the sphere is given by

$$\begin{aligned} V &= \frac{q}{\sqrt{r^2 + Z^2 - 2rZ \cos \theta}} + \frac{-q \frac{R}{Z}}{\sqrt{r^2 + \frac{R^4}{Z^2} - 2r \frac{R^2}{Z} \cos \theta}} \\ &= \boxed{\frac{q}{\sqrt{r^2 + Z^2 - 2rZ \cos \theta}} - \frac{q}{\sqrt{\left(\frac{rZ}{R}\right)^2 + R^2 - 2rZ \cos \theta}}} \end{aligned}$$

Now for the induced charge density:

$$\sigma = \frac{1}{4\pi} E_r = -\frac{1}{4\pi} \frac{\partial V}{\partial r}$$

Differentiate the formula above for the potential, and evaluate it at $r = R$:

$$\left. \frac{\partial V}{\partial r} \right|_{r=R} = q \left[\frac{-r + Z \cos \theta}{\left(r^2 + Z^2 - 2rZ \cos \theta \right)^{3/2}} - \frac{-\frac{rZ^2}{R^2} + Z \cos \theta}{\left(\left(\frac{rZ}{R} \right)^2 + R^2 - 2rZ \cos \theta \right)^{3/2}} \right]_{r=R}$$

$$= q \frac{-R + Z \cos \theta}{\left(R^2 + Z^2 - 2RZ \cos \theta \right)^{3/2}} - q \frac{-\frac{Z^2}{R} + Z \cos \theta}{\left(Z^2 + R^2 - 2RZ \cos \theta \right)^{3/2}}$$

$$= \frac{q}{R} \frac{Z^2 - R^2}{\left(R^2 + Z^2 - 2RZ \cos \theta \right)^{3/2}}$$

$$\sigma = -\frac{1}{4\pi} \left. \frac{\partial V}{\partial r} \right|_{r=R} = \boxed{-\frac{q}{4\pi R} \frac{Z^2 - R^2}{\left(R^2 + Z^2 - 2RZ \cos \theta \right)^{3/2}}}$$

Example - 2

From this we get the total charge induced on the grounded sphere:

$$\begin{aligned} Q &= \int \sigma R^2 \sin\theta \, d\theta \, d\phi = -\frac{q}{2} R (Z^2 - R^2) \int_0^\pi \frac{\sin\theta \, d\theta}{(R^2 + Z^2 - 2RZ \cos\theta)^{3/2}} \\ &= -\frac{q}{2} R (Z^2 - R^2) \left[\frac{-\frac{1}{ZR}}{\sqrt{R^2 + Z^2 - 2RZ \cos\theta}} \right]_0^\pi \\ &= \frac{q}{2} \frac{Z^2 - R^2}{Z} \left[\frac{1}{|R+Z|} - \frac{1}{|R-Z|} \right] = \boxed{-q \frac{R}{Z}} \end{aligned}$$

Goes to zero as Z
goes to infinity.

Example - 2

Potential energy: first, the force between the charge and image,

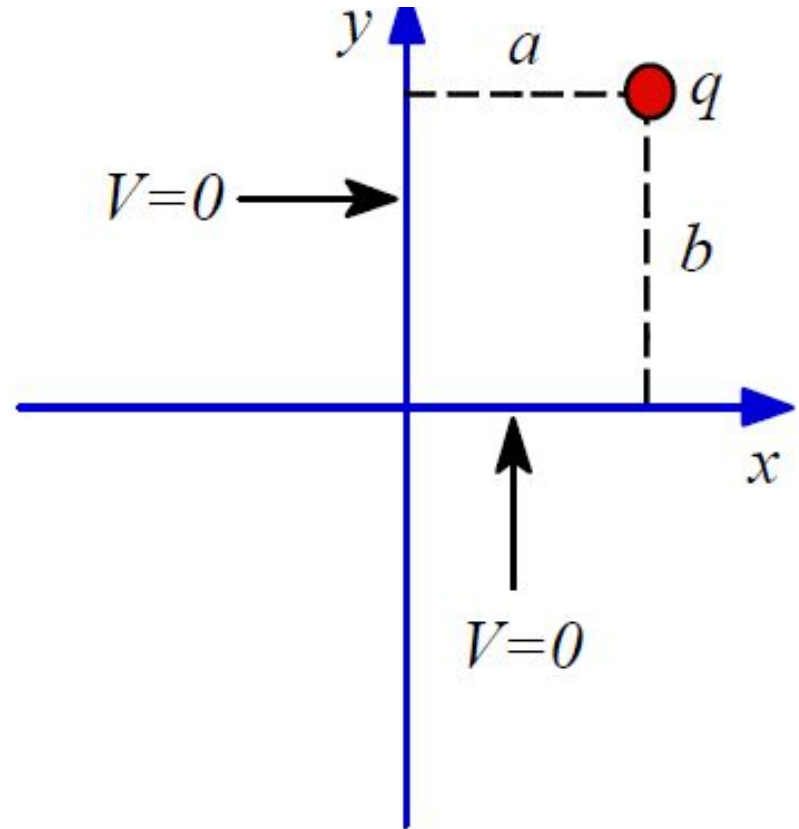
$$\mathbf{F}_{qq'} = \frac{qq'}{(Z - z')^2} \hat{\mathbf{z}} = \frac{q\left(-\frac{R}{Z}q\right)}{\left(Z - \frac{R^2}{Z}\right)^2} \hat{\mathbf{z}} = -\frac{ZRq^2}{(Z^2 - R^2)^2} \hat{\mathbf{z}}$$

and then the work to bring q in from infinity:

$$W = \int_{\infty}^Z -\mathbf{F}_{qq'} \cdot d\mathbf{l} = \int_{\infty}^Z \frac{zRq^2}{(z^2 - R^2)^2} dz = \frac{-Rq^2}{2(z^2 - R^2)} \Bigg|_{\infty}^Z = \boxed{-\frac{1}{2} \frac{Rq^2}{Z^2 - R^2}}$$

Example: Multiple Images

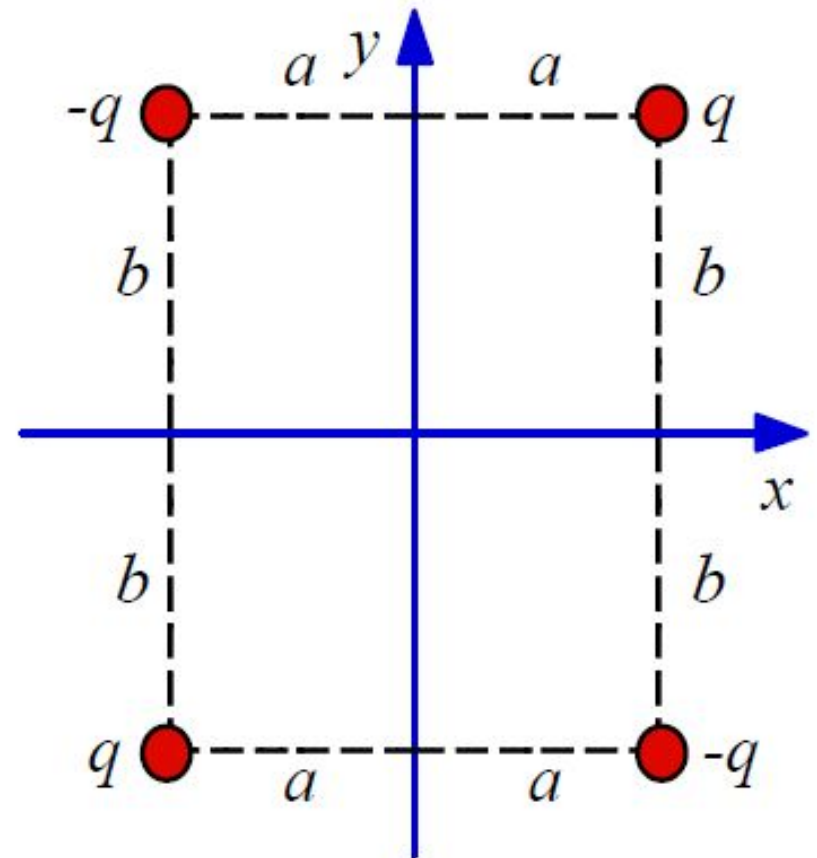
Two semi-infinite grounded conducting planes meet at right angles. In the region between them, there is a point charge q , situated as shown at right. Set up the image configuration, and calculate the potential in this region. What charges do you need, and where should they be located? What is the force on q ? How much work did it take to bring q in from infinity?



Hint:

In order to satisfy the boundary conditions, *three* image charges must be added to the system.

- ❑ The net force on q can be calculated by determining the vector sum of the forces on q due to the three image charges.
- ❑ The electrostatic energy of the real system is equal to $1/4$ of the electrostatic energy of the image-charge system.



Electric Fields in Matter

Polarization

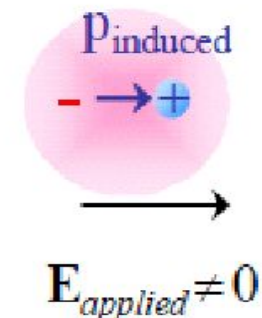
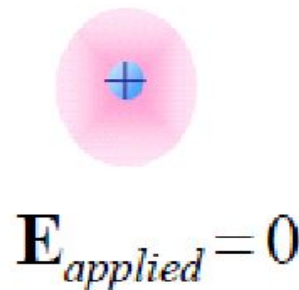
Dielectrics

All charges are attached to specific atoms or molecules.

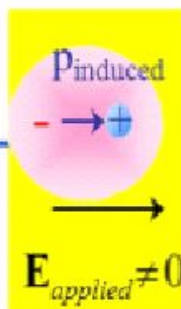
- They can move a bit *within* the atom or molecule.
- Such microscopic displacements account for the characteristic behavior of dielectric materials.

There are actually two principal mechanisms by which electric fields can distort the charge distribution of a dielectric atom or molecule:

→ *stretching and rotating.*



What happens to a neutral atom when it is placed in an electric field E ?



The nucleus is pushed in the direction of the field, and the electrons the opposite way. The two opposing forces reach a balance, leaving the atom **polarized**. The atom now has a tiny **dipole moment p** , which points in the *same direction* as E . Typically, this induced dipole moment is approximately proportional to the field (as long as the latter is not too strong):

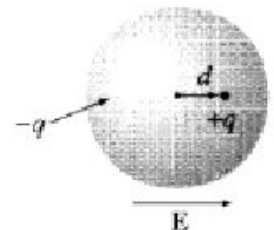
$$\boxed{p = \alpha E} \quad : \alpha \text{ is called } \mathbf{atomic\ polarizability}$$

Atomic Polarizabilities ($\alpha/4\pi\epsilon_0$, in units of 10^{-30} m^3)

H	He	Li	Be	C	Ne	Na	Ar	K	Cs
0.667	0.205	24.3	5.60	1.76	0.396	24.1	1.64	43.4	59.6

Example 4.1 An atom consists of a point nucleus ($+q$) surrounded by a uniformly charged spherical cloud ($-q$) of radius a . Calculate the atomic polarizability of such an atom.

It is reasonable to assume that the electron cloud retains its spherical shape. Equilibrium occurs when the nucleus is displaced a distance d from the center of the sphere and pulling by the internal field produced by the electron cloud, E_e :
At equilibrium, $E = E_e$



$$E = E_e = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} \longrightarrow p = qd = (4\pi\epsilon_0 a^3)E \longrightarrow \alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$$

Polarization

What happens to a piece of dielectric material when it is placed in an electric field?

*A lot of little dipoles pointing along the direction of the field.
The material becomes polarized.*

$$\mathbf{P} = \lim_{\Delta v \rightarrow 0} \left[\sum_i \mathbf{p}_i / \Delta v \right] \text{ (C/m}^2\text{)} \quad : \text{ Dipole moment per unit volume} \\ \rightarrow \text{Polarization}$$

This Polarization vector will displace the electric strength in a dielectric medium:

$$\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P} \quad : \mathbf{D} \text{ is known as the electric displacement}$$

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \quad : \text{For linear dielectrics}$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 (1 + \chi_e) \mathbf{E}$$

$$\mathbf{D} = \epsilon \mathbf{E} \quad \epsilon \equiv \epsilon_0 (1 + \chi_e) \quad : \text{Permittivity}$$

$$\epsilon_r \equiv 1 + \chi_e = \frac{\epsilon}{\epsilon_0} \quad : \text{Relative permittivity}$$

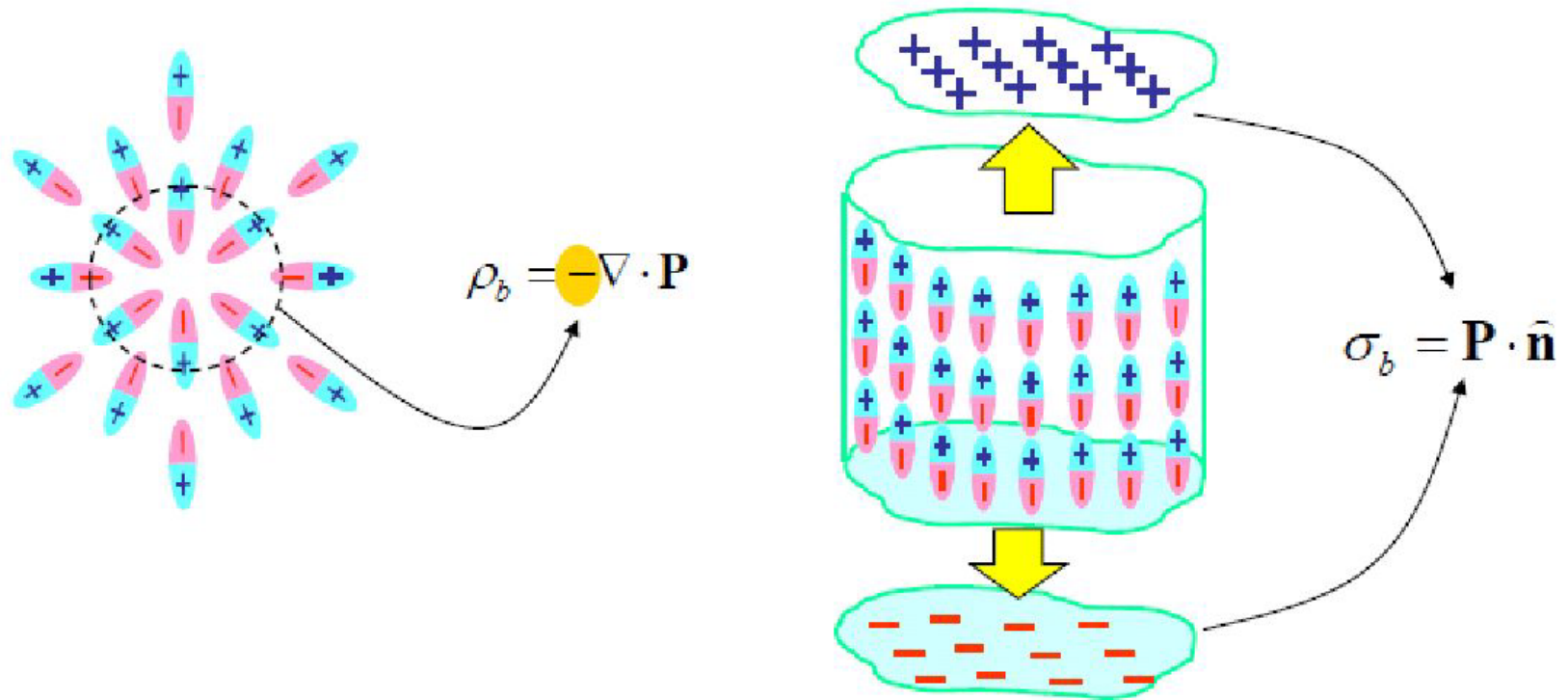
The Field of a Polarized Object

What is the field produced by $\mathbf{P}$ in a polarized material?

(not the field that may have caused the polarization, but the field the polarization itself causes)

Bound Charges

The polarization $\mathbf{P}$ can produce *bound volume and surface charges*.



Bound charge density

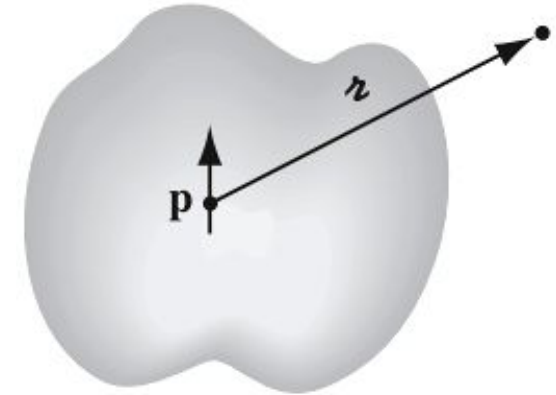
- ✓ Suppose we have a **piece of polarized material**, i.e., an object containing a **lot of microscopic dipoles lined up**.
- ✓ The **dipole moment per unit volume $\mathbf{P}$** is given.

▪ **Question: What is the field produced by this object?**

□ For a single dipole $\mathbf{p}$,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$

where $\mathbf{r}$ is the vector from the dipole to the point at which we are evaluating the potential.



We have a dipole moment $\mathbf{p} = \mathbf{P}d\tau'$ in each volume element $d\tau'$, so the **total potential** is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{r}}}{r^2} d\tau'$$

A little sleight-of-hand casts this **integral into a much more illuminating form**, observing that

$$\nabla' \left(\frac{1}{r} \right) = \frac{\hat{\mathbf{r}}}{r^2}$$

Bound charge density

Where the differentiation is with respect to the source coordinates ($\mathbf{r}'$), we have

$$V = \frac{1}{4\pi\epsilon_0} \int_V \mathbf{P} \cdot \nabla' \left(\frac{1}{r} \right) d\tau'$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\nabla' \left(\frac{1}{r} \right) = \frac{\hat{\mathbf{r}}}{r^2}$$

Integrating by parts

$$V = \frac{1}{4\pi\epsilon_0} \left[\int_V \nabla' \cdot \left(\frac{\mathbf{P}}{r} \right) d\tau' - \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau' \right]$$

or, invoking the [divergence theorem](#):

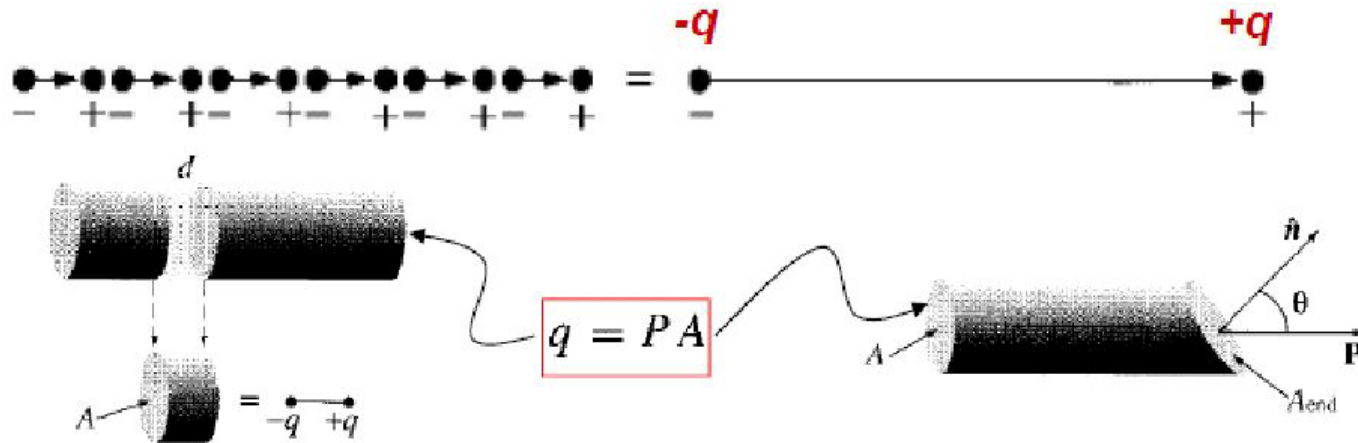
$$V = \frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{r} \mathbf{P} \cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau'$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{r} da' + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{r} d\tau'$$

$\rho_b \equiv -\nabla \cdot \mathbf{P}$: Volume charge density

$\sigma_b \equiv \mathbf{P} \cdot \hat{\mathbf{n}}$: Surface charge density

Physical Interpretation of Bound Charges



If the ends have been sliced off perpendicularly,

$$\sigma_b = \frac{q}{A} = P$$

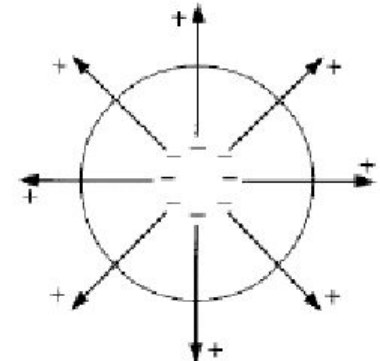
For an oblique cut, $A = A_{\text{end}} \cos \theta$,

$$\sigma_b = \frac{q}{A_{\text{end}}} = P \cos \theta = \mathbf{P} \cdot \hat{\mathbf{n}}$$

If the polarization is nonuniform, we get accumulations of bound charge *within* the material.

The net bound charge $\int \rho_b d\tau$ in a given volume is equal and opposite to the amount that has been pushed out through the surface:

$$\int_V \rho_b d\tau = - \oint_S \mathbf{P} \cdot d\mathbf{a} = - \int_V (\nabla \cdot \mathbf{P}) d\tau$$



Example

Problem: A dielectric cube of side $a = 2$ units, centered at the origin, carries a polarization $\mathbf{P} = 3\mathbf{r}$. Find the total volume bound charge and surface bound charge densities?

Solution:

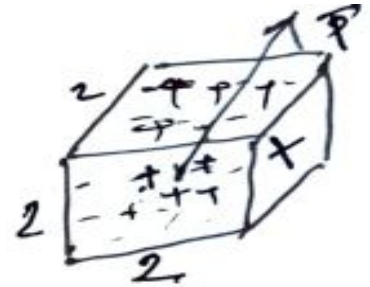
Volume charge density $\rho_b = -\nabla \cdot \mathbf{P}$ &

Surface charge density $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$

$$\mathbf{P} = 3\mathbf{r} = (3x)\hat{\mathbf{x}} + (3y)\hat{\mathbf{y}} + (3z)\hat{\mathbf{z}}$$

Volume bound charge density is

$$\rho_b = -\nabla \cdot \mathbf{P} = -3 \left[\left(\frac{\partial x}{\partial x} \right) \hat{\mathbf{x}} + \left(\frac{\partial y}{\partial y} \right) \hat{\mathbf{y}} + \left(\frac{\partial z}{\partial z} \right) \hat{\mathbf{z}} \right] = -3 \times 3 = -9$$



Total volume bound charge, $Q_{bV} = V \cdot \rho_b = -9 \times 2^3 = -72 \text{ units}$

Surface bound charge density, $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$

At the top surface, $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ and $z = a/2$

$$\text{Hence, } \sigma_{b\text{top}} = \hat{\mathbf{z}} \cdot \left[(3x)\hat{\mathbf{x}} + (3y)\hat{\mathbf{y}} + \left(\frac{3a}{2} \right) \hat{\mathbf{z}} \right] = \frac{3a}{2} = 3 \text{ (since, } a = 2 \text{ units)}$$

Similarly for all 6 surfaces, $\sigma_b = 6 \times 3 = 18$

Total surface bound charge, $Q_{bS} = S \cdot \sigma_b = 2^2 \times 18 = 72 \text{ units}$

Gauss's law in the presence of dielectrics

The effect of polarization ($\mathbf{P}$) is to produce accumulations of bound charge:

$$\rho_b \equiv -\nabla \cdot \mathbf{P} \quad : \text{within the dielectric} \quad \sigma_b \equiv \mathbf{P} \cdot \hat{\mathbf{n}} \quad : \text{on the surface}$$

If any free charge exists within a dielectric, therefore, the total charge density can be written as

$$\rho = \rho_b + \rho_f$$

→ Gauss's law reads

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_b + \rho_f = -\nabla \cdot \mathbf{P} + \rho_f \longrightarrow \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_f$$

$$\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P} \quad (C / m^2) \quad \rightarrow \text{Electric displacement}$$

$$\nabla \cdot \mathbf{D} = \rho_f$$

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{fenc}$$

Q_{fenc} denotes the total free charge enclosed in the volume.

In a typical problem, we know ρ_f , but we do not (initially) know ρ_b

Whenever the requisite symmetry is present, we can immediately calculate D by the Gauss's law method, because it makes reference only to free charges.

Similar to Gauss's law

Example

Problem 4.15 A thick **spherical shell** (inner radius **a**, outer radius **b**) is made of dielectric material with a "frozen-in" polarization **P(r)**. (No free charge inside)

Find the electric field in all three regions **by two different methods**:

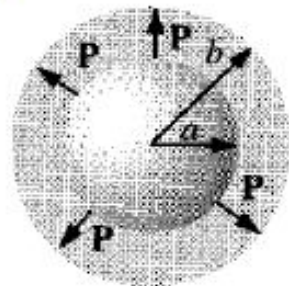
$$\mathbf{P}(\mathbf{r}) = \frac{k}{r} \hat{\mathbf{r}}$$

(a) Identify all the bound charge, and use Gauss's law to calculate the field it produces.

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{k}{r} \right) = -\frac{k}{r^2};$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = \begin{cases} +\mathbf{P} \cdot \hat{\mathbf{r}} = k/b & (\text{at } r = b), \\ -\mathbf{P} \cdot \hat{\mathbf{r}} = -k/a & (\text{at } r = a). \end{cases}$$

$$\mathbf{P}(\mathbf{r}) = \frac{k}{r} \hat{\mathbf{r}}$$



For $r < a$, $Q_{\text{enc}} = 0$, so Gauss's law $\Rightarrow \mathbf{E} = 0$.

$$\text{For } r > b, \quad Q_{\text{tot}} = \oint_S \sigma_b da + \int_V \rho_b d\tau = \oint_S \mathbf{P} \cdot d\mathbf{a} - \int_V \nabla \cdot \mathbf{P} d\tau$$

(divergence theorem)

$$\oint_S \mathbf{P} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{P} d\tau, \text{ so } Q_{\text{enc}} = 0 \quad \text{Gauss's law} \Rightarrow \mathbf{E} = 0$$

$$\text{For } a < r < b, \quad Q_{\text{enc}} = \left(\frac{-k}{a} \right) (4\pi a^2) + \int_a^r \left(\frac{-k}{\bar{r}^2} \right) 4\pi \bar{r}^2 d\bar{r} = -4\pi ka - 4\pi k(r - a) = -4\pi kr; \text{ so}$$

$$\mathbf{E} = -(k/\epsilon_0 r) \hat{\mathbf{r}}.$$

(b) Use the equations of electric displacement **D**: $\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}$

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f_{\text{enc}}} = 0 \Rightarrow \mathbf{D} = 0 \text{ everywhere.} \quad \mathbf{E} = (-1/\epsilon_0) \mathbf{P}, \text{ so}$$

$$\mathbf{E} = 0 \text{ (for } r < a \text{ and } r > b);$$

$$\mathbf{E} = -(k/\epsilon_0 r) \hat{\mathbf{r}} \text{ (for } a < r < b)$$



*Much simpler to use **D**.*

Example

Problem: Find the magnitudes of **D** and **P** for a dielectric material in which $E=0.15$ MV/m and $\chi_e = 4.25$. Here, $\epsilon_0=8.85\times 10^{-12}$ F/m

Solution:

Here, $E = 0.15$ MV/m

$$\chi_e = 4.25$$

Vacuum permittivity, $\epsilon_0 = 8.85\times 10^{-12}$
F/m

We know that $\epsilon_r = 1 + \chi_e = 5.25$

$$\begin{aligned} \text{i) } \mathbf{D} &= \epsilon_0 \epsilon_r E \\ &= (8.85\times 10^{-12}) (5.25) (0.15\times 10^6) \text{ (F/m)(V/m)} \end{aligned}$$

$$\mathbf{D} = 6.96 \mu\text{C/m}^2 \quad [\text{Since } F=C/V]$$

$$\begin{aligned} \text{ii) } \mathbf{P} &= \epsilon_0 \chi_e E \\ &= (8.85\times 10^{-12}) (4.25) (0.15\times 10^6) \text{ (F/m)(V/m)} \end{aligned}$$

$$\mathbf{P} = 5.64 \mu\text{C/m}^2$$

Boundary Conditions

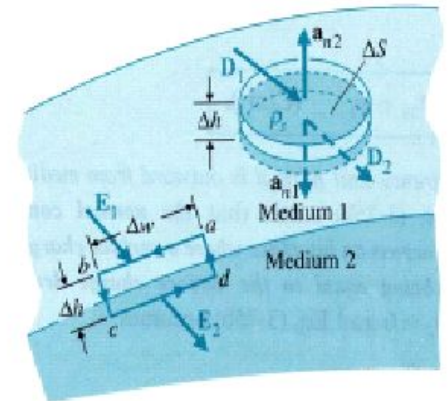
$$\nabla \cdot \mathbf{D} = \rho_f \rightarrow \oint_S \mathbf{D} \cdot d\mathbf{a} = (\mathbf{D}_1 \cdot \mathbf{a}_{n2} + \mathbf{D}_2 \cdot \mathbf{a}_{n1}) \Delta S = \mathbf{a}_{n2} \cdot (\mathbf{D}_1 - \mathbf{D}_2) \Delta S = (D_{1n} - D_{2n}) \Delta S$$

$$\rightarrow \int_V \rho_f dv = (\rho_f \Delta h) \Delta S$$

$$\Rightarrow \boxed{D_{1n} - D_{2n} = \sigma_f} \quad \sigma_f = \lim_{\Delta h \rightarrow 0} \rho_f \Delta h$$

$$\nabla \times \mathbf{D} = \nabla \times \mathbf{P} \rightarrow \oint_C \mathbf{D} \cdot d\mathbf{l} = (\mathbf{D}_1 \cdot \mathbf{l}_2 + \mathbf{D}_2 \cdot \mathbf{l}_1) = \mathbf{l}_2 \cdot (\mathbf{D}_1 - \mathbf{D}_2) = (D_{1t} - D_{2t})$$

$$\Rightarrow \boxed{D_{1t} - D_{2t} = P_{1t} - P_{2t}}$$



Note for E:

$$E_{1n} - E_{2n} = \frac{\sigma_{enclosed}}{\epsilon_0} = \frac{\sigma_f + \sigma_b}{\epsilon_0}$$

$$\boxed{E_{1t} - E_{2t} = 0}$$

Summary of Boundary conditions

$$\mathbf{E}_{above}^{\parallel} - \mathbf{E}_{below}^{\parallel} = 0 \quad \mathbf{D}_{above}^{\parallel} - \mathbf{D}_{below}^{\parallel} = \mathbf{P}_{above}^{\parallel} - \mathbf{P}_{below}^{\parallel}$$

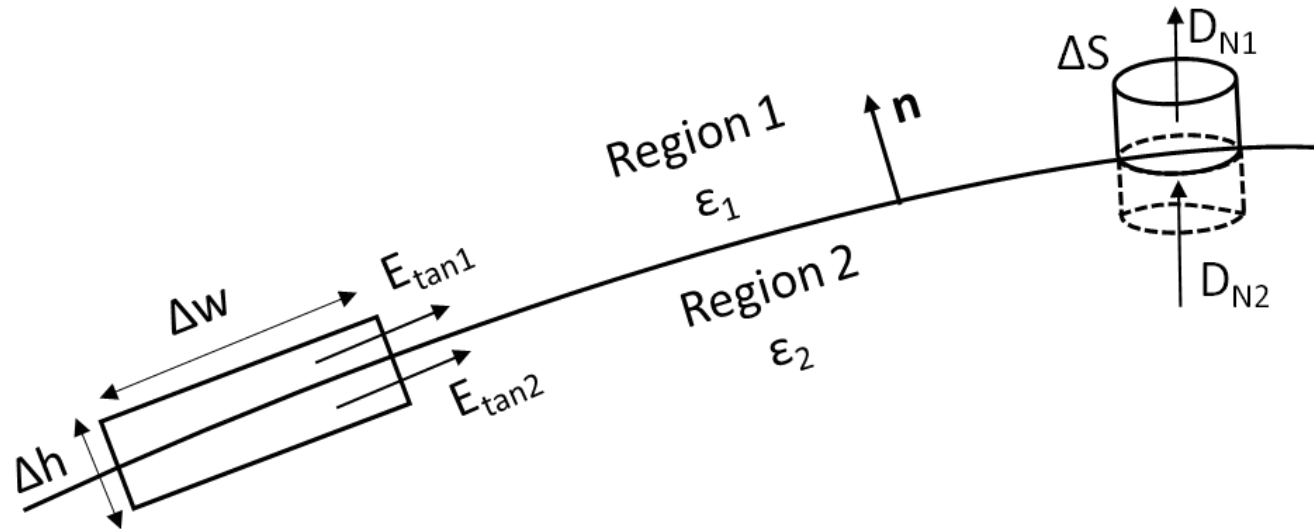
$$E_{above}^{\perp} - E_{below}^{\perp} = \frac{1}{\epsilon_0} \sigma \quad D_{above}^{\perp} - D_{below}^{\perp} = \sigma_f$$

$$(\sigma = \sigma_f + \sigma_b)$$

Example

Problem: Two dielectrics with the dielectric constant of 5 and 12 are separated by an interface at the plane $x=5$. The electric field ($\mathbf{E}$) is given for one side of the interface (with a dielectric constant of 5) as $\mathbf{E}_1 = 36\hat{x} - 16\hat{y} + 20\hat{z} \text{ V/m}$. Evaluate the $\mathbf{E}$ on the other side of the interface.

Solution:



Given, $\mathbf{E}_1 = 36\hat{x} - 16\hat{y} + 20\hat{z} \text{ V/m}$, and $\epsilon_1 = 5\epsilon_0$

The tangential component by using, $\oint \mathbf{E} \cdot d\mathbf{L} = 0$

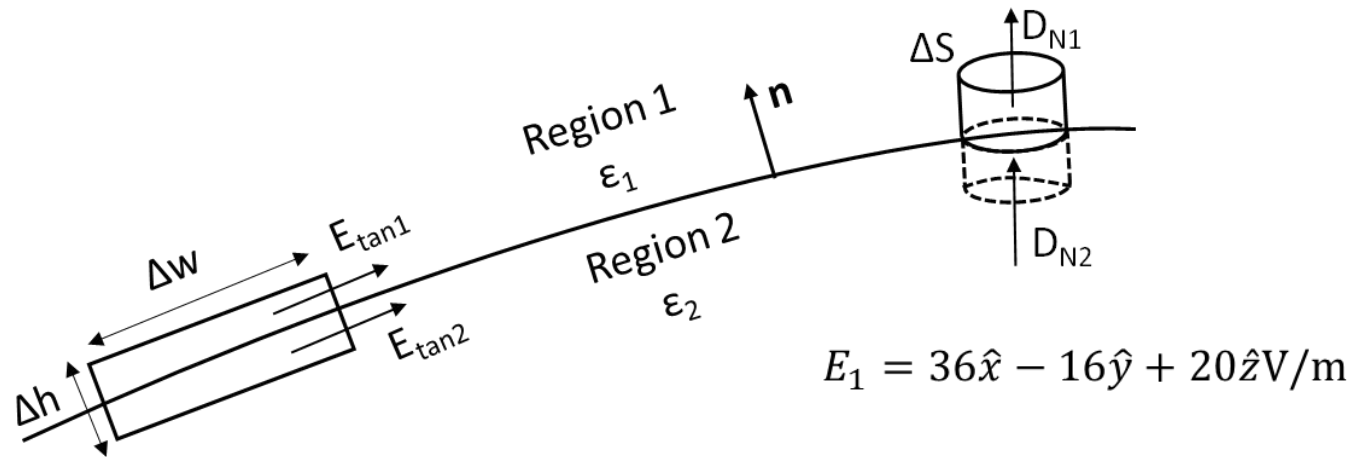
Around the small closed path on the left, obtaining, $E_{\tan 1} \Delta w - E_{\tan 2} \Delta w = 0$

So, $E_{\tan 1} = E_{\tan 2}$ or, $E_{1t} = E_{2t}$

Therefore, $E_{2y} = -16 \text{ V/m}$ & $E_{2z} = 20 \text{ V/m}$

Cont.

The boundary conditions on the normal components are found by applying Gauss's law to the small "pillbox" shown on the right side of the figure.



The sides are again **very short** & the **flux leaving the top and bottom surface** is

$$D_{N1}\Delta S - D_{N2}\Delta S = \Delta Q = \rho_s \Delta S$$

From which, $D_{N1} - D_{N2} = \rho_s$

Assuming, ρ_s is zero, $D_{N1} - D_{N2} = 0 \Rightarrow D_{N1} = D_{N2}$

$$\Rightarrow \epsilon_1 E_{1x} = \epsilon_2 E_{2x}$$

$$\text{Therefore, } E_{2x} = \frac{\epsilon_1}{\epsilon_2} E_{1x} = \frac{5}{12} \times 36$$

Since, $E_{2y} = -16 \text{ V/m}$ & $E_{2z} = 20 \text{ V/m}$ $= 15 \text{ V/m}$

$$\text{Thus, } E_2 = 15\hat{x} - 16\hat{y} + 20\hat{z} \text{ V/m}$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

$$\text{or } \Phi_E = \oint \vec{D} \cdot d\vec{s} = Q_{enc}$$

$$\text{Here, } D = \epsilon_0 E$$

Dielectric susceptibility

- We know that the polarization of a dielectric normally results from an electric field, which lines up the atomic or molecular dipoles.
- For many substances, in fact, the polarization is proportional to the field, provided E is not too strong,

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

- ✓ The constant of proportionality, χ_e , is called the electric susceptibility of the medium.
- ✓ The value of χ_e depends on the microscopic structure of the substance and also on external conditions such as temperature.
- We shall call materials that obey $\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$ as linear dielectrics.
- Note that $\mathbf{E}$ in above equation is the total field.
- It may be due in part to free charges and in part to the polarization itself.
- If, we put a piece of dielectric into an external field $\mathbf{E}_0$, we cannot compute $\mathbf{P}$ directly from above equation.
- Since, the external field will polarize the material, and this polarization will produce its own field, which then contributes to the total field, and this in turn modifies the polarization.

Dielectric susceptibility

□ In **linear media**, we have

$$\begin{aligned}\mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} \\ &= \epsilon_0 (1 + \chi_e) \mathbf{E}\end{aligned}\quad \boxed{\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}}$$

□ So $\mathbf{D}$ is also proportional to $\mathbf{E}$: $\mathbf{D} = \epsilon \mathbf{E}$, where $\epsilon \equiv \epsilon_0 (1 + \chi_e)$

□ This **new constant** ϵ is called the **permittivity of the material**.

- In vacuum, where there is no matter to polarize, the **susceptibility is zero**, and the **permittivity is ϵ_0** .
- Therefore, ϵ_0 is called the **permittivity of free space**.

□ If we remove a factor of ϵ_0 , the remaining is dimensionless quantity

$\epsilon_r \equiv 1 + \chi_e = \epsilon / \epsilon_0$ is called the **relative permittivity**, or **dielectric constant**, of the material.

- ϵ_r is **greater than 1**, for all ordinary materials.

Example

Problem: The dielectric material has a relative dielectric constant of 2.5 and a uniform flux density of 6 nC/m^2 . Find the (i) electric field (E), (ii) polarization (P), and (iii) average number of dipoles (n) per cubic meter? Here, the average dipole moment, p: 10^{-30} C.m .

Solution:

$$\text{Relative dielectric constant}(\epsilon_r) = 2.5$$

$$\text{Average flux density}(D) = 6 \text{ nC.m}^{-2}$$

$$\text{Average dipole moment}(p) = 10^{-30} \text{ C.m}$$

$$D = \epsilon E$$

$$E = \frac{D}{\epsilon} \quad [\epsilon = \epsilon_0 \epsilon_r]$$

$$\text{Here, } \epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

$$\epsilon = \epsilon_0 \epsilon_r = 8.85 \times 10^{-12} \times 2.5$$

$$= 22.125 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

$$\text{(i) Electric field } E = \frac{D}{\epsilon}$$

$$= \frac{6 \text{ nC/m}^2}{22.125 \times 10^{-12} \text{ C}^2/\text{N.m}^2}$$

$$\begin{aligned} \text{Electric Field}(E) &= 0.27118 \times 10^{12} \text{ nN/C} \\ &= 271.18 \text{ N/C} \end{aligned}$$

Example

Problem: Cont... (ii) polarization (P), and (iii) average number of dipoles (n) per cubic meter?
Here, the average dipole moment, p: 10^{-30} C.m?

Solution: cont...

Relative dielectric constant (ϵ_r) = 2.5

Average flux density(D) = $6nC.m^{-2}$

(ii) Polarization (P) = $\chi_e \epsilon_0 E$

Here, $\chi_e = \epsilon_r - 1 \Rightarrow 2.5 - 1 = 1.5$

$E = 271.18N/C$ and $\epsilon_0 = 8.85 \times 10^{-12}C^2/N.m^2$

Therefore, $P = (1.5)(8.85 \times 10^{-12})(271.18).C/m^2$

$P = 5399.87 \times 10^{-12}.C/m^2$

$P \sim 5.4nC/m^2$

Relation between the polarization(P) and dipole moment(p)

$P = np$ (here, n is average number of dipoles)

(iii) Avg. no of dipoles = $n = \frac{P}{p}$

$= \frac{5.4nC/m^2}{10^{-30}C.m}$
 $n = 5.4 \times 10^{21}m^{-3}$

Example

Problem: A metal sphere of radius a carries a charge Q . It is surrounded, out to radius b , by linear dielectric material of permittivity ϵ . Find the potential at the center (relative to infinity)?

Solution: To compute V , we need to know E

To find E , we might first try to locate the bound charge

We could get the bound charge from P , but we can't calculate P unless we already know E .

We know that the free charge is Q , and the arrangement is spherically symmetric, so let's begin by calculating D

$$\mathbf{D} = \frac{Q}{4\pi r^2} \hat{\mathbf{r}}$$

for all points $r > a$

Inside the metal sphere can be $\mathbf{E} = \mathbf{P} = \mathbf{D} = 0$

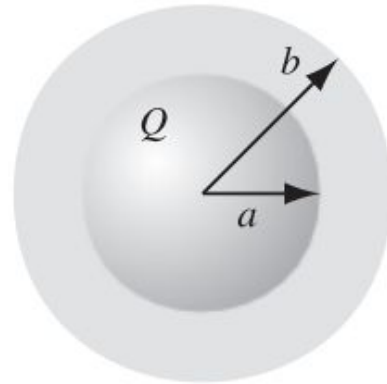
Once we know D , it is a trivial matter to obtain E

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{E} = \mathbf{D} / \epsilon$$



$$\mathbf{E} = \begin{cases} \frac{Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}, & \text{for } a < r < b, \\ \frac{Q}{4\pi \epsilon_0 r^2} \hat{\mathbf{r}}, & \text{for } r > b. \end{cases}$$



Example

The **potential at the center** is therefore,

$$V = - \int_{\infty}^0 \mathbf{E} \cdot d\mathbf{l} = - \int_{\infty}^b \left(\frac{Q}{4\pi\epsilon_0 r^2} \right) dr - \int_b^a \left(\frac{Q}{4\pi\epsilon r^2} \right) dr - \int_a^0 (0) dr$$

$$= \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b} \right).$$

$$V = \frac{Q}{4\pi} \left(\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b} \right)$$

$$\mathbf{E} = \begin{cases} \frac{Q}{4\pi\epsilon r^2} \hat{\mathbf{r}}, & \text{for } a < r < b, \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, & \text{for } r > b. \end{cases}$$

Further, the **polarization** in the dielectric material can be

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} = \frac{\epsilon_0 \chi_e Q}{4\pi \epsilon r^2} \hat{\mathbf{r}}$$

Energy in Dielectric Systems

It takes **work** to charge up a capacitor: $W = (1/2)CV^2$

If the **capacitor (C)** is filled with **linear dielectric**, its capacitance **exceeds the vacuum value (C_{vac})** by a factor of the dielectric constant (ϵ_r), $C = \epsilon_r C_{vac}$

Obviously, the work necessary to charge a **dielectric-filled capacitor** is **increased** by the **same factor**.

The reason is so clear: You have to **pump on more (free) charge**, to **achieve a given potential**, because *part of the field is canceled off by the bound charges*.

A general formula for the **energy stored in any electrostatic system**:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

The case of the **dielectric-filled capacitor** suggests that this should be changed to

$$W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Since, $D = \epsilon E$

in the **presence of linear dielectrics**

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C}$$

or, since $Q = CV$,

$$W = \frac{1}{2} CV^2,$$

where V is the final potential of the capacitor.

Example

Problem: Calculate the total energy for a concentric linear dielectric material having radius R_1 and susceptibility χ_e by surrounding with a spherical conductor of radius R_2 , which carries a charge 'q'.

Solution:

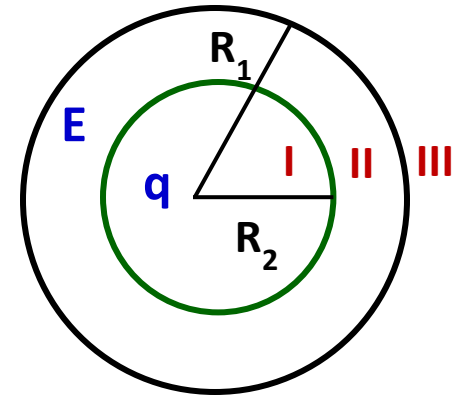
$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

For $r < R_2$ $\vec{D} = 0$ then $\vec{E} = 0$

For $r > R_2$ then $\vec{D} = \frac{q}{4\pi r^2} \hat{r}$

$\vec{E} = \frac{q}{4\pi\epsilon_0\epsilon_r r^2} \hat{r}$ for $R_2 < r < R_1$

and $\vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$ for $r > R_1$



Total energy:

$$\begin{aligned} W &= \frac{1}{2} \int_v \vec{D} \cdot \vec{E} d\tau \\ &= 0 + \frac{1}{2} \int_{R_1}^{R_2} \frac{q}{4\pi\epsilon_0\epsilon_r r^2} dr + \frac{1}{2} \int_{R_2}^{\infty} \frac{q}{4\pi\epsilon_0\epsilon_r r^2} dr \\ &= \frac{q^2}{8\pi\epsilon_0} \left[\frac{1}{\epsilon_r} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{1}{R_2} \right] \end{aligned}$$

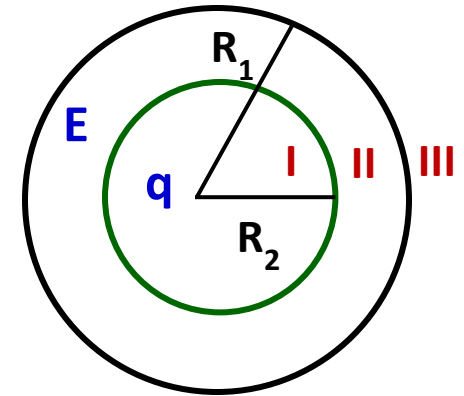
Example

Problem: Calculate the total energy for a concentric linear dielectric material.....

Cont.....

Total energy:

$$\begin{aligned}
 W &= \frac{1}{2} \int_v \vec{D} \cdot \vec{E} d\tau \\
 &= 0 + \frac{1}{2} \int_{R_1}^{R_2} \frac{q}{4\pi\epsilon_0\epsilon_r r^2} dr + \frac{1}{2} \int_{R_2}^{\infty} \frac{q}{4\pi\epsilon_0\epsilon_r r^2} dr \\
 &= \frac{q^2}{8\pi\epsilon_0} \left[\frac{1}{\epsilon_r} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) + \frac{1}{R_2} \right] \\
 &= \frac{q^2}{8\pi\epsilon_0\epsilon_r} \left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{\epsilon_r}{R_2} \right] \\
 &= \frac{q^2}{8\pi\epsilon_0\epsilon_r} \left[\frac{R_2 - R_1 + \epsilon_r R_1}{R_1 R_2} \right], \text{ since } \epsilon_r = 1 + \chi_e \\
 &= \frac{q^2}{8\pi\epsilon_0(1 + \chi_e)} \left[\frac{R_2 - R_1 + (1 + \chi_e)R_1}{R_1 R_2} \right] \\
 W &= \frac{q^2}{8\pi\epsilon_0(1 + \chi_e)} \left[\frac{1}{R_1} + \frac{\chi_e}{R_2} \right]
 \end{aligned}$$



Example

Problem: A metallic sphere of radius 10 cm has a surface charge density of 10 nC/m^2 . Calculate the **electric energy stored in the system**.

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Solution: Metallic sphere, $r=10 \text{ cm}=10 \times 10^{-2} \text{ m}$ and $\sigma=10 \text{ nC/m}^2=10 \times 10^{-9} \text{ C/m}^2$

Energy stored in system,
$$W = \frac{1}{2} \int V \cdot \sigma ds = \frac{1}{2} Q \cdot V$$

Where, Q is the total charge

Now
$$V = \int \frac{\sigma ds}{4\pi\epsilon_0 r} = \frac{\sigma}{4\pi\epsilon_0 r} \int_S r^2 \sin \theta d\theta d\phi$$

$$= \frac{10 \times 10^{-9}}{4\pi\epsilon_0} \cdot r \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi$$

$$= 9 \times [-\cos \theta]_0^\pi \times [\phi]_0^{2\pi}$$

$$= 113.1 \text{ V}$$

$$W = \frac{1}{2} Q \cdot V, Q = \sigma A = \sigma 4\pi r^2$$

$$Q = 10 \times 10^{-9} \times 4\pi \times (10 \times 10^{-2})^2$$

$$Q = 1.257 \times 10^{-9} \text{ C}$$

$$W = \frac{1}{2} \times 1.257 \times 10^{-9} \times 113.1$$

$$= 71.08 \times 10^{-9} \text{ J}$$

Example

Problem: When **two dielectrics** are present in a capacitor with the interface parallel to E and D, if the separation is 4 mm having areas of $A_1 = 0.2 \text{ m}^2$, $A_2 = 0.3 \text{ m}^2$ and $\epsilon_{r1} = 2.0$, $\epsilon_{r2} = 3.0$ of two dielectrics. **Find the equivalent capacitance?**

Solution:

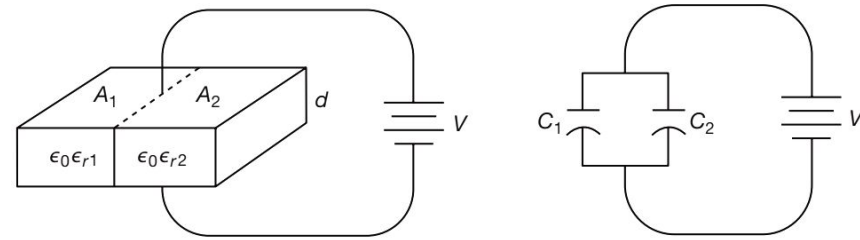
The equivalent capacitance can be obtained by treating the arrangement as two capacitors in parallel.

Here, $d = 4 \text{ mm}$, $A_1 = 0.2 \text{ m}^2$, $A_2 = 0.3 \text{ m}^2$, $\epsilon_{r1} = 2.0$ and $\epsilon_{r2} = 3.0$

The vacuum permittivity, $\epsilon_0 = 8.8541 \times 10^{-12} \text{ F/m}$

$$C_1 = \frac{\epsilon_0 \epsilon_{r1} A_1}{d} \text{ and } C_2 = \frac{\epsilon_0 \epsilon_{r2} A_2}{d}$$

$$\begin{aligned} C_{eq} &= C_1 + C_2 = \frac{\epsilon_0}{d} (\epsilon_{r1} A_1 + \epsilon_{r2} A_2) \\ &= \frac{8.8541 \times 10^{-12} \text{ F/m}}{4 \times 10^{-3} \text{ m}} [(2.0)(0.2) \text{ m}^2 + (3.0)(0.3) \text{ m}^2] \\ &= \frac{8.8541 \times 10^{-12} \text{ F}}{4 \times 10^{-3} \text{ m}^2} (1.3) \text{ m}^2 \\ &= 2.8775 \times 10^{-9} \text{ F} \\ &= 2.88 \text{ nF} \end{aligned}$$



Capacitance

$$C = \frac{Q}{\Delta V} = \frac{\text{Coulombs}}{\text{Volts}} = \text{Farads}$$

$$\begin{aligned} \text{Electric Field between flat plates} \quad E &= \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, & \text{Potential Difference between flat plates} \quad \Delta V &= Ed = \frac{Qd}{\epsilon_0 A} \end{aligned}$$

$$C = \frac{Q}{\Delta V} = \frac{Q}{Qd/\epsilon_0 A} = \frac{\epsilon_0 A}{d}$$

Capacitance depends on the Area of the plates and the distance between the plates